Container Ships · Design Manual: Sizing from Bays, Rows and Tiers · Student Version

2026-11-11 · T. Santos, L. Batista

A design manual for the preliminary sizing of a cellular container ship from the number of containers it must carry. It supplements the Ship Types deck on container ships and is the method behind the container-capacity work of the Ship Design Project.

About this manual

Purpose. A container ship is designed around its cargo: the number of containers fixes the grid of bays, rows and tiers, and the grid fixes the main dimensions. This manual turns that idea into a method you can follow step by step, with every formula, its source and a worked example on a real ship; Appendix D lists the Python functions that apply it and gives the code of the harder steps. It follows the design sequence of Alvariño, Azpíroz and Meizoso (1997), *El Proyecto Básico del Buque Mercante*, section 2.9.3, from the course bibliography, cited as Alvariño et al.; Papanikolaou (2026), the course textbook, is used to cross-check the results.

Scope. The manual covers the preliminary sizing of a gearless cellular container ship with its engine room aft:

- the split of the capacity between holds and deck, and first estimates of the dimensions from statistics;
- the grid of rows, tiers and bays in the holds, and the main dimensions built from it;
- the deck capacity under the sight line from the bridge;
- the block coefficient and draught, the weights, a first stability check, and the checks that close the design;
- how to arrive at several possible ships rather than a single answer, so that the best one can be chosen later in the project.

Limitations. The method is a first sizing tool and is kept simple on purpose. These simplifications limit what its results can say:

What the method simplifies	What it means for the results	What to keep in mind
The first estimates come from statistics of built ships of a limited range of sizes, which carried containers of standard height	ships that carry high-cube containers come out too shallow; very small or very large ships fall outside the data	compare the first estimates with other built ships; expect a deeper hull with high cubes
The capacity of the holds comes from an average loss factor, not from the shape of the hull	the number of bays, and so the length, can be off by about two bays, some 13 m	treat the length as a range until the hull shape is known
The deck is first counted with the same height of stacks along the whole ship	the deck capacity is overestimated at the bow, where the view from the bridge limits the stacks	reduce the forward stacks, as the method shows
No hull lines are drawn; the hull form is described by a few coefficients	the form coefficients are estimates, and there are no hydrostatic curves	confirm them once the lines are drawn

What the method simplifies	What it means for the results	What to keep in mind
The weights come from simple formulas, some of them fitted on larger ships	the empty weight of a small ship is uncertain	compare with a similar ship already built
Stability is only a first check	no damage stability, no large angles of heel, no lashing of the deck stacks	use it to discard ships, not to approve one
Only gearless ships with the engine room and bridge aft are treated	ships with their own cranes, or with the bridge forward, have a different length and a different view over the deck	adapt the method and state what was changed

How general the method is. The sequence itself does not depend on the size of the ship: estimate the dimensions from statistics, turn each estimate into the whole numbers of containers on either side of it, build the hull around the containers, and keep the ships that pass the checks.

What limits it are the numbers it is fed. The statistics cover a range of sizes and ships of a certain age; the share of containers in the holds changes with the size and service of the ship; and some allowances, such as the length of the engine room, are taken from one ship. For a ship unlike those behind the statistics, replace these numbers with data from similar built ships; the method stays the same. When a canal, a port or the cranes fix the beam, the beam is not estimated: the number of containers across the ship is fixed first, and only the height is varied.

Place in the course. The manual supplements the Ship Types lecture on container ships. It gives the method for the first sizing of the container ship in the Ship Design Project. Later work in the course refines these first results, for example by counting the containers in the actual shape of the hull.

How to use it.

- **First reading:** Part I (concepts) and the overview at the start of Part II give the whole picture.
- **Designing:** work through Step 0 · Inputs to Step 8 · Check and iterate in order. Each step opens with a card that lists its purpose, inputs, outputs, equations and code, and closes with a short list of checks before you move on.
- **Checking your numbers:** Part III runs the method on the reference ship, step by step, with every intermediate number.
- **Looking up a symbol:** Appendix A lists every symbol with its unit.

Conventions. Symbols are set in `code style`, for example `L_PP` or `n_R`. Units are metres, tonnes, knots and kW. The manual's own equations stand on their own lines, numbered (1) to (36) in order, and are cited as eq. (8). Equations of the sources keep their own numbers: Eq. 2.9.4 is equation 2.9.4 of Alvariño et al., § marks a section of their text, and app. 2.9 is their appendix 2.9 of statistics of built ships.

One numbering for the method: the steps of Part II, Step 0 · Inputs to Step 8 · Check and iterate, and their sections; 3.2 Beam is section 2 of Step 3 · Main dimensions from the container grid. Alvariño et al. number the same calculations differently, as boxes in their flow charts; those numbers appear only in Appendix C, which says where each box is done in this manual.

Contents

Part	Section	What it gives you
I · Concepts	I.1 The container and the cargo grid	TEU, ISO sizes, bays, rows, tiers, holds and deck

Part	Section	What it gives you
	I.2 The hull around the grid	hull parts, guide clearances, other clearances, main dimensions
	I.3 Hold utilization `k_U`	why the hull carries less than the full grid, and by how much
	I.4 Targets, weights and candidates	first estimates, displacement and its parts, the set of candidate ships
	I.5 The reference ship	the 596 TEU feeder used throughout
II · Method	The method at a glance	flowchart, the nine steps, where each variable comes from
	Step 0 · Inputs	what to collect before sizing
	Step 1 · Split and targets	eqs. (1)–(5)
	Step 2 · Container grids	eqs. (6)–(8), four grids per share
	Step 3 · Main dimensions	eqs. (9)–(11), ratio check
	Step 4 · Deck and visibility	eqs. (12)–(17)
	Step 5 · Displacement and form	eqs. (18)–(20)
	Step 6 · Weights	eqs. (21)–(30)
	Step 7 · Stability	eqs. (31)–(36)
	Step 8 · Checks	the five checks and the domain of candidates
III · Application	III.1 Worked example	the method run on the reference ship, the sweep, the `k_U` sensitivity
	III.2 Validation	the estimates against two built ships
Appendices	A Nomenclature	every symbol with its unit
	B Notes on the sources	where the manual departs from its sources, and why
	C Alvariño's boxes and the sections of this manual	the five blocks of Alvariño et al. on one page
	D Writing the code	what your program must do, the functions step by step, and the code of the harder steps
	E References	

Part I · Concepts

I.1 The container and the cargo grid

The container. Capacity is counted in TEU, twenty-foot equivalent units: one TEU is one 20 ft container. A 40 ft container, an FEU, counts as two TEU. The sizes are fixed by ISO 668:

Container	Length	Width	Height	TEU
20 ft	6.058 m	2.438 m	2.591 m (8 ft 6 in); 2.896 m (high cube)	1
40 ft (FEU)	12.192 m	2.438 m	2.591 m (8 ft 6 in); 2.896 m (high cube)	2
45 ft	13.716 m	2.438 m	2.896 m	not counted in TEU sizing

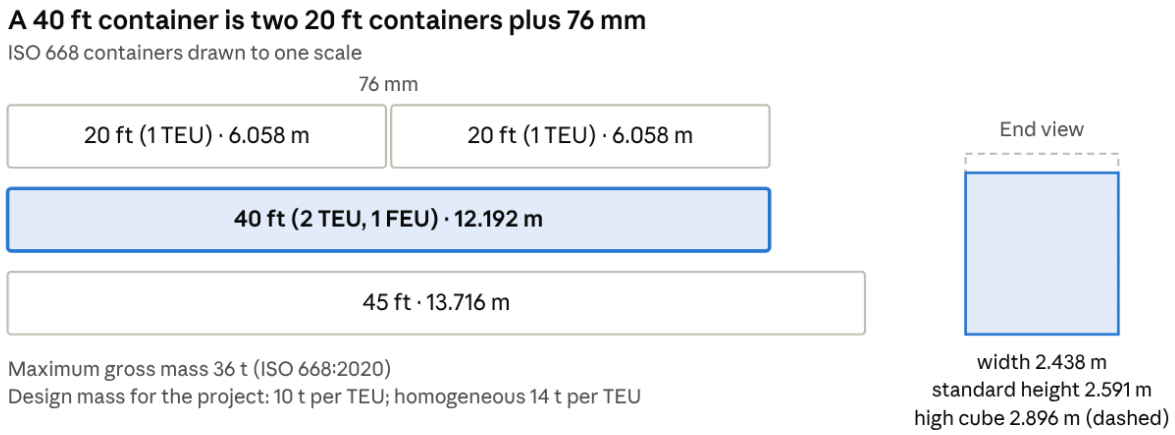


Figure I.1 · ISO 668 containers to one scale

A 40 ft container is two 20 ft containers plus the 76 mm gap between them (§2.9.2). The formulas of Alvariño et al. assume 8 ft 6 in containers (§3.8.6.1); this manual uses the same formulas with 2.896 m in place of 2.591 m when the ship carries high cubes.

The mass of a container. The mass of a container depends on what is inside it. ISO 668 allows a 20 ft container up to 36 t, but a ship never carries all its containers that heavy: it would sink far below its marks. For the weights, the design therefore works with an average mass per TEU, given by the owner. The Ship Design Project uses two values:

- **10 t per TEU, the design load.** The average mass of a normal cargo. The brief gives the cargo as `N_TEU` · 10 t, and the design loading conditions of Step 7 use it.
- **14 t per TEU, a heavy, even load.** Every container weighs the same, 14 t. It is used to check the deepest draught and the stability with heavy cargo. At this mass the ship usually carries fewer containers than its TEU capacity: its deadweight or its stability runs out before its slots do.

The grid. In the holds the containers stand in a grid of three whole numbers:

- `n_R` **rows** across the beam;
- `n_T` **tiers**, one above the other;
- `n_B` **bays** along the length, each bay one 20 ft container long.

Each position in the grid is a **cell**, or slot. The containers of one row in one bay form a **stack**. A **hold** is the space between two transverse bulkheads, the watertight walls across the ship; it takes two 20 ft bays, or one 40 ft bay, so `n_B` is normally even and the number of holds is `n_H0 = n_B / 2`.

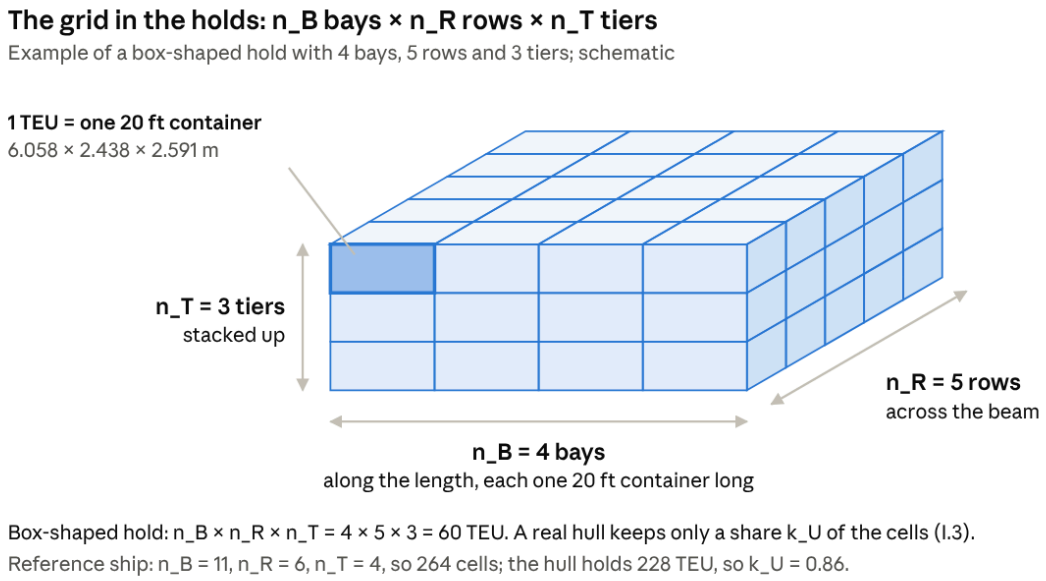


Figure I.2 · The grid in the holds: bays, rows and tiers

Holds and deck. The three integers count the holds only. On deck the containers stand on the hatch covers that close the holds, usually two rows wider than in the hold, with their own number of tiers `n_TD`. The owner asks for a total capacity `N_TEU`; the design splits it into `N_H` TEU in the holds and `N_D` on deck, and `s_H = N_H / N_TEU` is the share in the holds. The two parts are limited by different things, which is why the split is the first decision of the method:

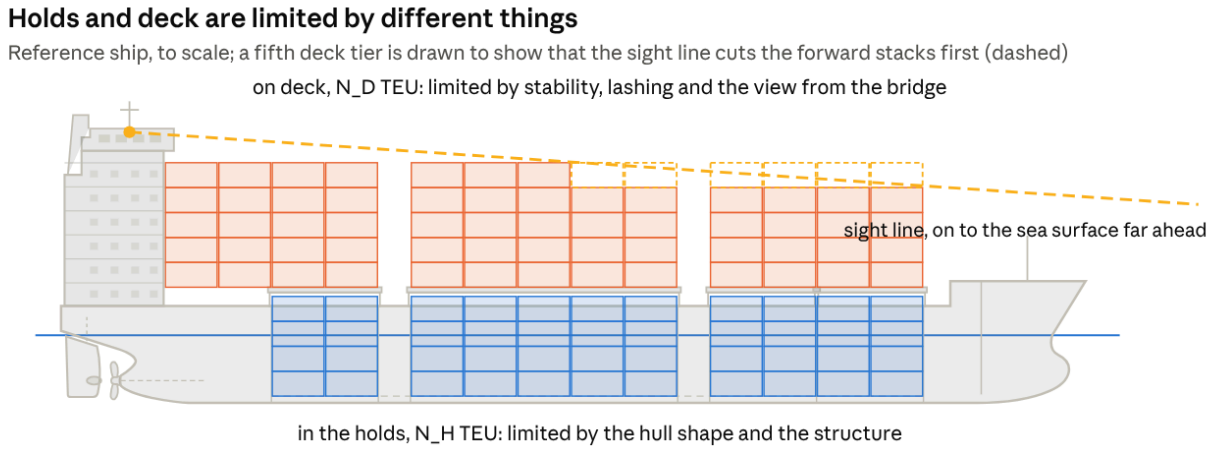


Figure I.3 · Holds and deck, and what limits each

- **In the holds** the hull limits the capacity: the beam and depth set the rows and tiers, and the hull narrows at the ends.
- **On deck** stability, lashing and the view from the bridge limit the capacity: the stacks raise the centre of gravity, must be secured against the ship's motions, and must stay below the sight line that SOLAS V/22 requires from the bridge.

I.2 The hull around the grid

The main dimensions are the containers plus the structure around them. The space each part takes is an **allowance** added to the containers. This section lists every allowance once, with its typical value; the

steps only refer back to it.

Hull parts.

Part	What it is	Symbol	Typical value	Source
Double bottom	the space between the bottom shell and the inner bottom, under the lowest tier	`h_DB`	1.5 m to a little over 2 m	§2.9.4
Double hull, or wing tanks	the space between the side shell and the inner side of the hold	`w_DH`	at least 2.0–2.5 m; about 0.1 `B`, rarely over 2.5 m	§2.9.3.1, §2.9.4
Hatch coaming	the frame around the hatch opening, rising above the deck	`h_CO`	from a reference ship	–
Hatch cover	the lid that closes the hatch; the deck containers stand on it	`t_cover`	from a reference ship	–
Margin under the hatch cover	the space between the top tier in the hold and the underside of the hatch cover	`c_HC`	4–5 % of the smaller hatch-cover dimension	§2.9.3.1
Deck camber	the rise of the deck at the centreline	`c_DK`	from a reference ship; 0 for a flat deck	–
Transverse bulkhead	the watertight wall between two holds; the length it takes	`t_BH`	from a reference ship	–
Aft peak, engine room, fore peak	spaces at the stern, for the machinery and at the bow, with no containers in the hold	`L_AP`, `L_ER`, `L_FP`	from similar ships	§2.9.3.1
Deckhouse	the superstructure with the accommodation and the navigation bridge; the eye of the officer on watch sets the sight line over the deck stacks	–	from a reference ship	–

Guide clearances. Cell guides are vertical steel angles that the containers slide down into; they hold each stack in place against the ship's motions. Each guide takes some space, so a clearance is added at every guide. The first letter after `g_` gives the direction, L along the ship or B across it; the second gives where the guide stands:

Clearance	Where	Typical value	Source
`g_LB`	lengthwise, between the end of a container and the transverse bulkhead, where the guides are fixed to the bulkhead	0.115–0.120 m	Table 2.9.2
`g_LF`	lengthwise, between the two 20 ft containers of one hold, where the guides stand on their own	0.180–0.200 m	Table 2.9.2
`g_BC`	across, between the outer row and the side of the hold, under the hatch coaming	0.150–0.600 m	Table 2.9.2
`g_BF`	across, between two neighbouring rows, where the guides stand between them	0.080–0.100 m	Table 2.9.2

So a hold is two containers long plus `2 g_LB + g_LF`, and the cargo space is `n_R` containers wide plus `2 g_BC + (n_R − 1) g_BF`.

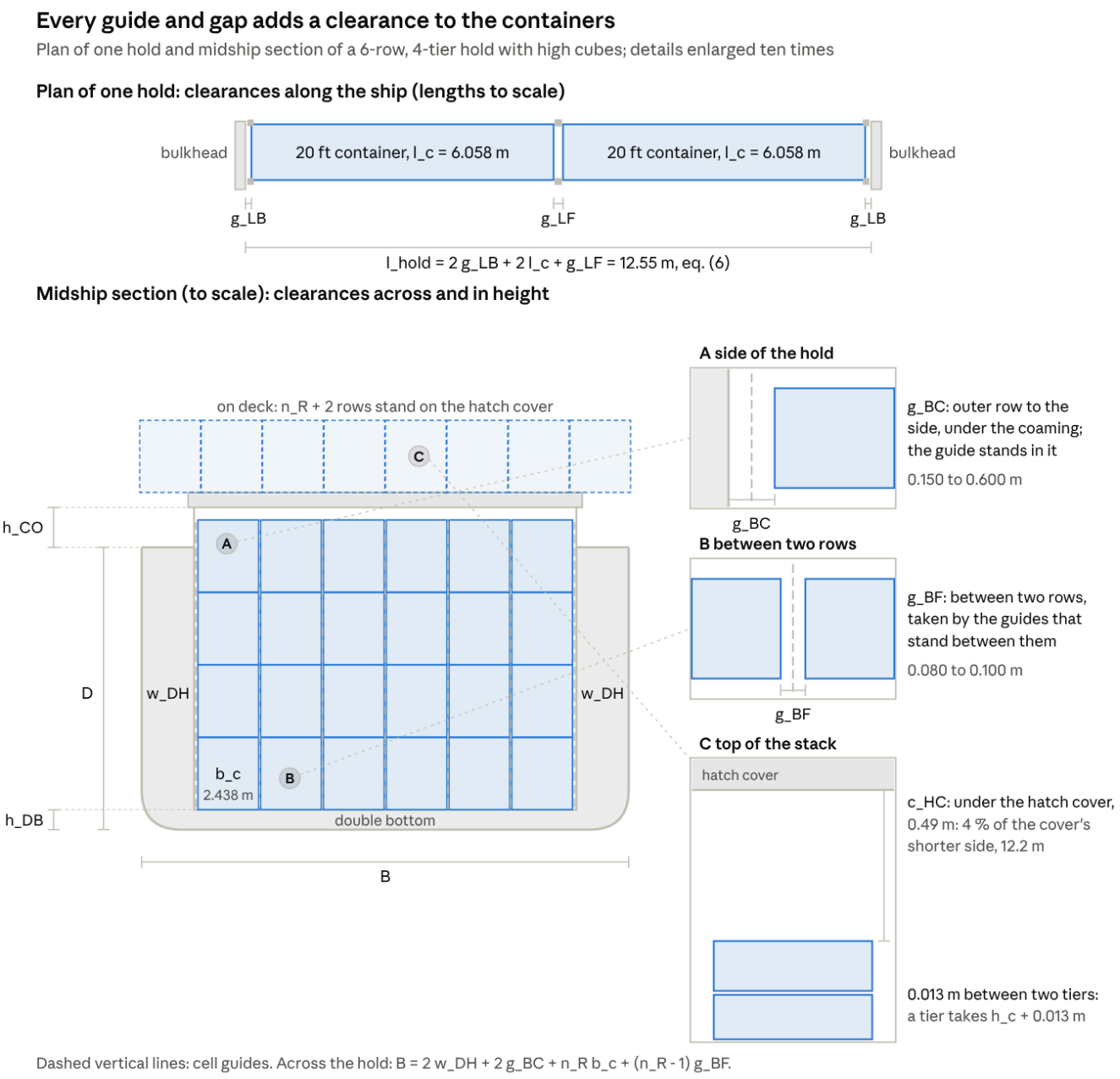


Figure I.4 · Guide clearances and gaps: plan of one hold and midship section

Other gaps and pitches. Besides the guides, two gaps are added. For a first estimate, each container can simply be given a fixed space, called its pitch: the distance from one container to the next, container and gaps included:

Item	Value	Source
Gap between two stacked containers	0.013 m, so each tier takes `h_c` + 0.013 m	§3.8.6.1
Gap between two 40 ft containers lengthwise	0.9 m	§3.8.6.1
Row pitch in the hold, first estimate	2.54 m: the container plus 100 mm for the guides	Eq. 3.8.30
Tier pitch	2.604 m for 8 ft 6 in containers; 2.909 m for high cubes	Eq. 3.8.31

Item	Value	Source
Bay pitch, first estimate	6.55 m per 20 ft bay: half a 40 ft container plus half the 0.9 m gap	Eq. 3.8.32
Row pitch on deck	2.464 m	Eq. 3.8.34

Use the values of a reference ship whenever you have them; the table gives what to use when you do not.

Main dimensions and reference lines. `L_PP` is the length between perpendiculars: from the aft perpendicular (AP), at the rudder stock, to the forward perpendicular (FP), where the design waterline meets the stem. `L_OA` is the length overall, from the stern to the bow. Heights are measured from the base line, at the keel. `D` is the depth, from the base line to the deck at the side; `B` is the beam; `T` is the draught, from the base line to the waterline. The freeboard is the height of the deck above the waterline.

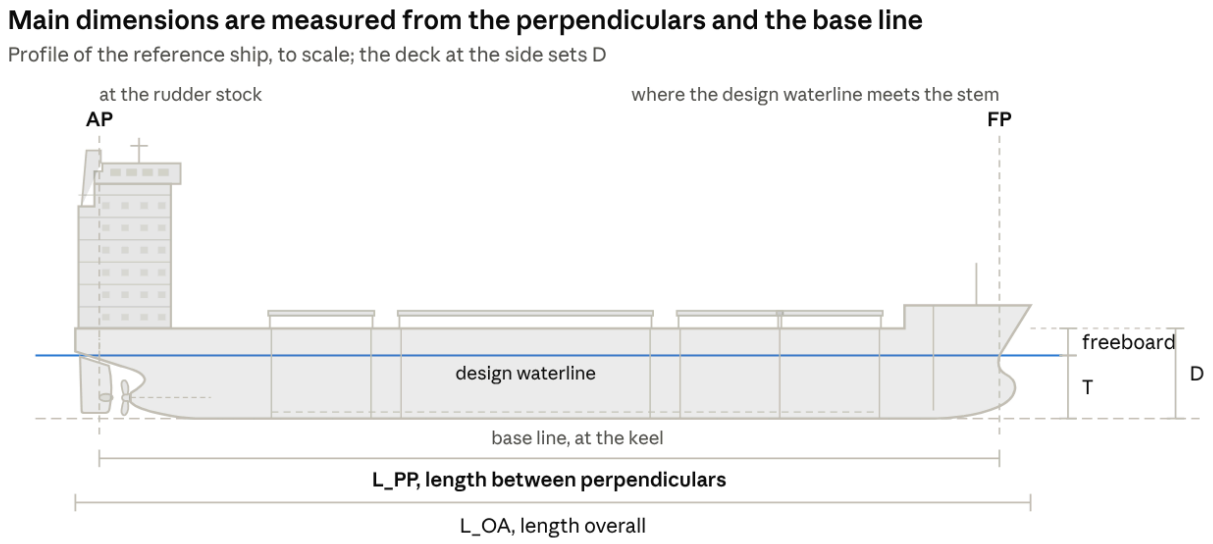


Figure I.5 · Perpendiculars, base line and the main dimensions

I.3 Hold utilization `k_U`

What it is. The product `n_B · n_R · n_T` counts the containers of a box-shaped hold. A real ship carries fewer. The hull narrows towards the bow and the stern and its bottom rises at the ends, so the end bays take fewer rows and tiers; and any cell that a tank, a space or structure takes away is lost as well. The hold utilization `k_U` is the share of the full grid that the ship really carries:

`k_U = N_H / (n_B · n_R · n_T)`

A `k_U` measured on a built ship includes all of these losses together; it cannot tell how many come from the hull shape and how many from the inside arrangement.

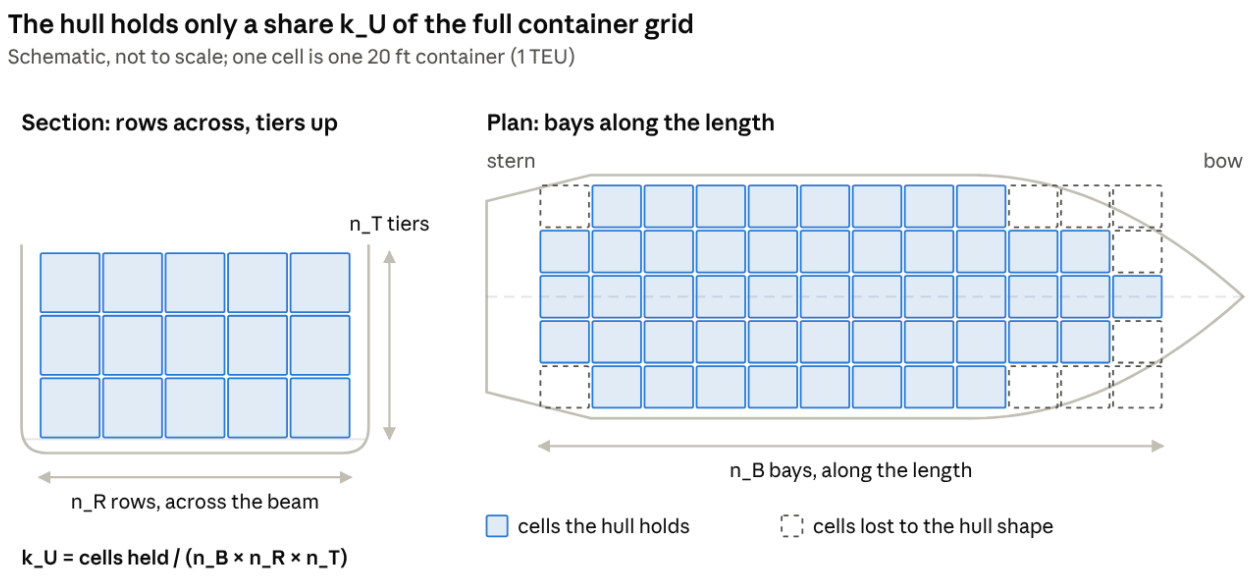


Figure I.6 · The hull holds a share k_U of the full grid (schematic)

Why it matters. `k_U` links the TEU in the holds to the number of bays: there must be enough bays that `k_U · n_B · n_R · n_T ≥ N_H`, eq. (8) of Step 2 · Lay out the containers. So `k_U` sets the length: the smaller it is, the more bays the ship needs. Two bays more or less change the length by about 13 m.

How it is estimated. When there is no statistic from similar ships, Alvariño et al. give `k_U = 0.5 C_B + 0.4` (Eq. 2.9.4): a fuller hull loses less space at the ends. `C_B` is known only once the draught is solved, in Step 5 · Displacement, C_B and draught, so the first pass uses an assumed `C_B` and is repeated with the computed one (Step 5 · Displacement, C_B and draught, 5.4 Close the iteration with step 2).

How good the formula is. On the reference ship (I.5), the assumed `C_B` = 0.68 gives `k_U` = 0.5 × 0.68 + 0.4 = 0.74. The real ship holds 228 TEU in a grid of 11 × 6 × 4 = 264 cells, so its `k_U` is 0.86: the formula is 14 % low. To hold 238 TEU in 6 rows and 4 tiers, 0.74 needs 14 bays and 0.86 only 12.

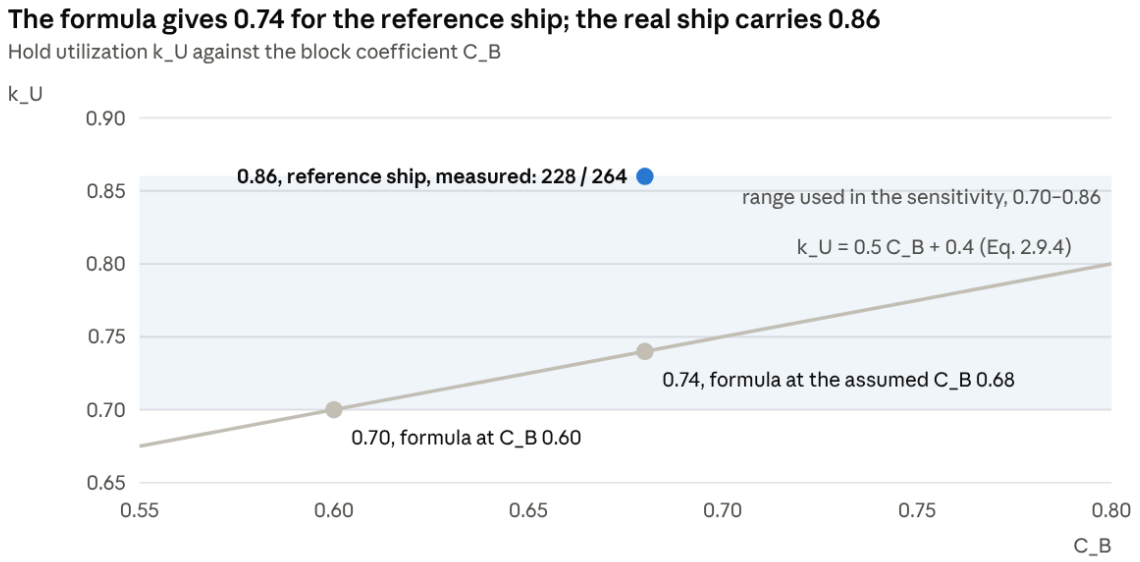


Figure I.7 · k_U by Eq. 2.9.4 and on the reference ship; C_B 0.68 assumed

How the manual treats it. `k_U` is not a design choice like the share in the holds or the grid: it follows from the hull of each candidate, and the formula only estimates it. So it is not swept; it is iterated with `C_B` and its uncertainty is shown as a sensitivity, with bounds taken from the data at hand:

Bound	Value	Where it comes from
Low	0.70	the formula at `C_B` = 0.591, the lowest `C_B` that Step 5 · Displacement, C_B and draught gives for the reference ship (at 18 kn), rounded
Formula	0.74	the formula at the assumed `C_B` = 0.68
High	0.86	measured on the reference ship

Whenever a reference ship's hold capacity and grid are known, measure its `k_U` and use it instead of the formula. Once the hull lines are drawn, the containers can be counted directly in the shape of the hull, and the factor is no longer needed.

I.4 Targets, weights and candidates

Targets. Statistics of built ships give first estimates of the length, beam, depth and draught from the TEU in the holds: `L_0`, `B_0`, `D_0` and `T_0`. They are **targets**, marked with the subscript 0, not the dimensions of the design. The main dimensions `L_PP`, `B` and `D` are built afterwards from whole rows, tiers and bays plus the allowances of I.2.

The draught `T_0` of the statistics is the freeboard draught of the built ships, the draught at their summer load line. It serves for the ratios of Step 3 · Main dimensions from the container grid and for comparison; it is not a limit on the draught of the design.

Weights and form.

Term	Symbol	Meaning
Displacement	`Δ`	the mass of the ship, equal to the mass of the water it displaces
Lightship	`LWT`	the empty ship: steel, machinery and outfit
Deadweight	`DWT`	everything the ship carries: cargo, fuel, water, stores and crew; `Δ = LWT + DWT`
Block coefficient	`C_B`	the underwater volume over `L_PP · B · T`; a full hull has a high `C_B`
Midship and prismatic coefficients	`C_M`, `C_P`	the midship section area over `B · T`; `C_P = C_B / C_M`
Service speed	`V`	in knots; with the length it sets the Froude number `Fn`

Candidates. The owner fixes how many containers the ship must carry, but not where they go. More containers in the holds need a larger hull; more containers on deck need a stable ship and a clear view from the bridge over the stacks. There is no single right answer, so the method does not choose one split: it tries several, from about a third to a half of the containers in the holds.

The estimate falls between whole numbers. For each split, statistics of built ships give a first estimate of the beam and the depth. But a real ship carries **whole rows and whole tiers** of containers, so the estimate almost always falls **between two whole numbers**: between 6 and 7 rows, say, or between 3 and 4 tiers.

Because the statistics scatter, there is no reason to prefer the lower number, so the method keeps both: the lower and the higher number of rows, each with the lower and the higher number of tiers. One more row gives a wider and shorter hull; one more tier, a deeper and shorter hull. Each combination of a split and an arrangement gives a different ship, a **candidate**.

Simple checks of proportions, capacity, weight and stability then remove the candidates that do not work. The candidates left form the set of acceptable designs, and the project chooses among them later, on cost and performance.

How many designs. The numbers stay small and easy to follow. The method tries five splits, from about a third to a half of the containers in the holds in steps of 5 %, and four arrangements for each split: 5 × 4 = **20 candidates**. If a canal or a port fixes the beam, only the height is varied, and there are 5 × 2 = 10.

Every candidate goes through the same calculations and the same checks. Its weights are balanced by iteration: the draught is adjusted until the hull floats what it carries. A candidate is dropped only when something cannot be fixed that way: its proportions are wrong, its deck containers need more than five tiers, the draught that carries the weights breaks the freeboard or the proportions, or it is not stable enough even with ballast.

The checks thin the list step by step. For the brief of the reference ship, 3 of the 20 pass Step 3 · Main dimensions from the container grid and Step 4 · Capacity on deck and visibility, and one of them passes all the checks of Step 8 · Check and iterate (III.1). The candidates that pass every check are the acceptable designs that go on to the economic evaluation, which chooses one.

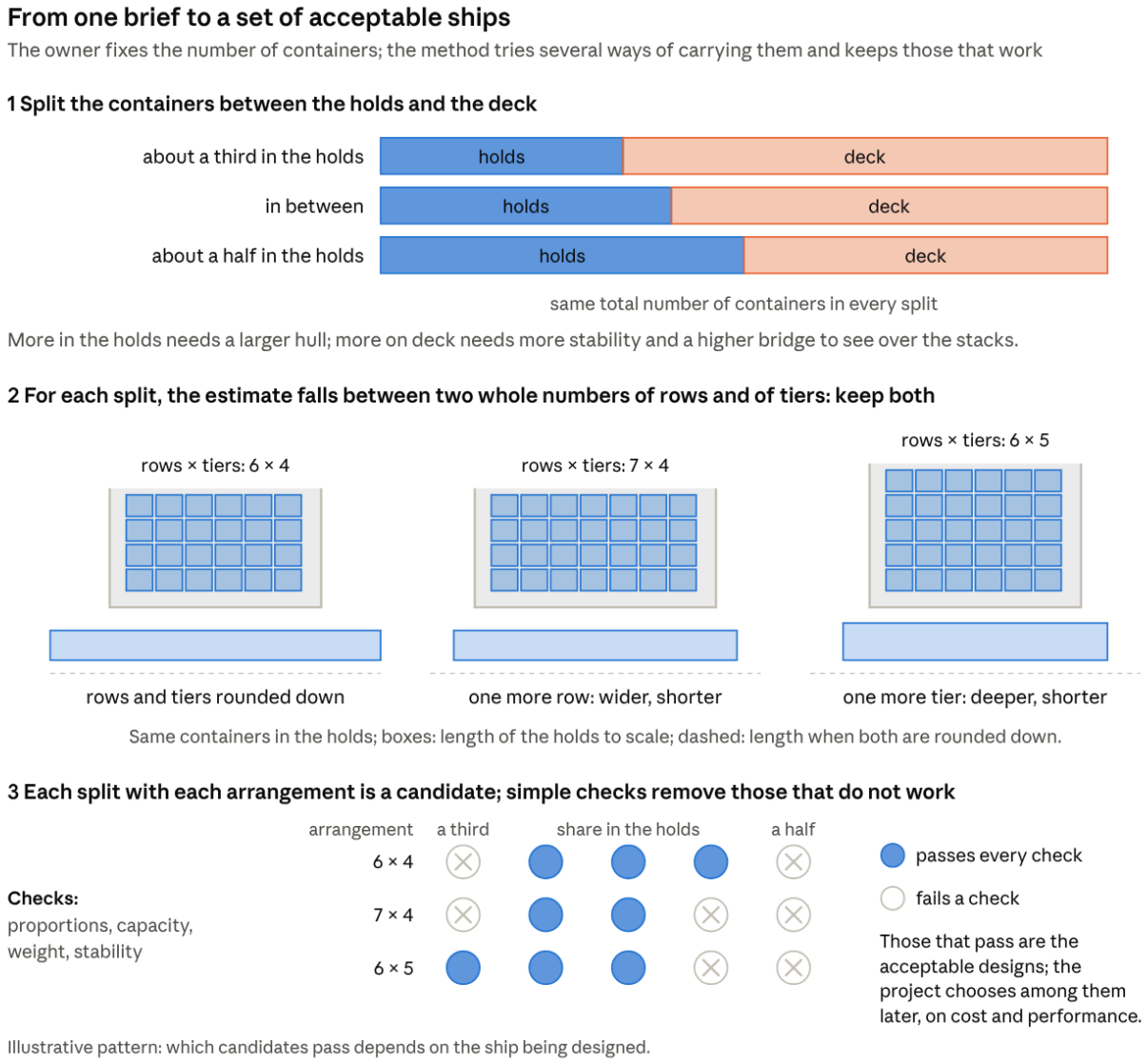


Figure I.8 · From one brief to a set of acceptable ships

I.5 The reference ship

A **reference ship** is a ship already built, of a size and type close to the one being designed. It gives the data that statistics cannot, such as the allowances of I.2, the lengths of the engine room and peaks, and `k_U`, and it is a yardstick for the results.

This manual uses one reference ship throughout: a feeder container ship of 596 TEU, built in 2007, with the engine room and bridge aft. Unlike the gearless ship that the method sizes, it is a **geared** ship: it has two cranes on the port side, and their pedestals take space between the holds. Keep this difference in mind when you use its data. Part III designs it again from its capacity and compares the result with what was built.

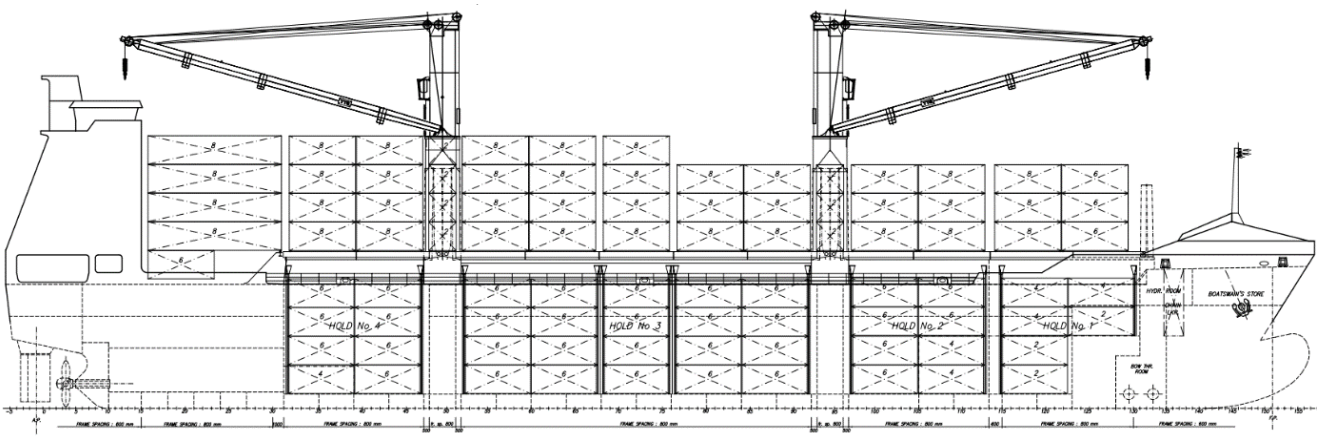


Figure I.9 · General arrangement of the reference ship: profile with holds 1 to 4, the deck stacks and the two cranes

Particular	Value
Type	geared feeder container ship, built 2007; engine room and bridge aft; two cranes of 45 t on the port side
Capacity	596 TEU counted on the general arrangement: 228 in the holds (`s_H` = 0.38), 368 on deck. The published nominal capacity is 620 TEU; at 14 t each, 410 TEU
Main dimensions	`L_OA` 119.8 m, `L_PP` 112.9 m, `B` 20.0 m, `D` 11.3 m, `T` 7.9 m (summer draught)
Deadweight	8,893 t at the summer draught
Gross tonnage	7,064
Machinery	one medium-speed diesel engine, 6,000 kW at 500 rpm
Service speed	16.5 kn
Containers	high cube, 2.896 m high
Holds	4 holds with 11 bays of 20 ft (2, 2 + 1 + 2, 2 and 2 bays); 6 rows; 4 tiers
Deck	13 bays (11 over the holds, 2 over the engine room); 8 rows; 4 tiers
Length budget	aft peak and engine room 21.6 m; cargo zone 76.2 m (holds 68.7 m, spaces between holds 7.5 m, where the cranes stand); fore peak 15.1 m
Allowances	double bottom 0.8 m; wing tanks 2.1 m; coaming 1.6 m; hatch cover 0.6 m; outer deck rows on a 0.15 m side support
Deckhouse	aft, over the engine room, 12 m long

Particular	Value
Hold utilization	`k_U` = 228 / (11 × 6 × 4) = 0.86
Not published	lightship, block coefficient, midship coefficient and eye height; where a value is needed, this manual assumes the eye 1.8 m above the bridge deck, `C_B` = 0.68 and `C_M` = 0.98

The main dimensions, deadweight, tonnage, machinery, speed and cranes are the ship's published particulars; the grid, the allowances and the length budget are measured on its general arrangement.

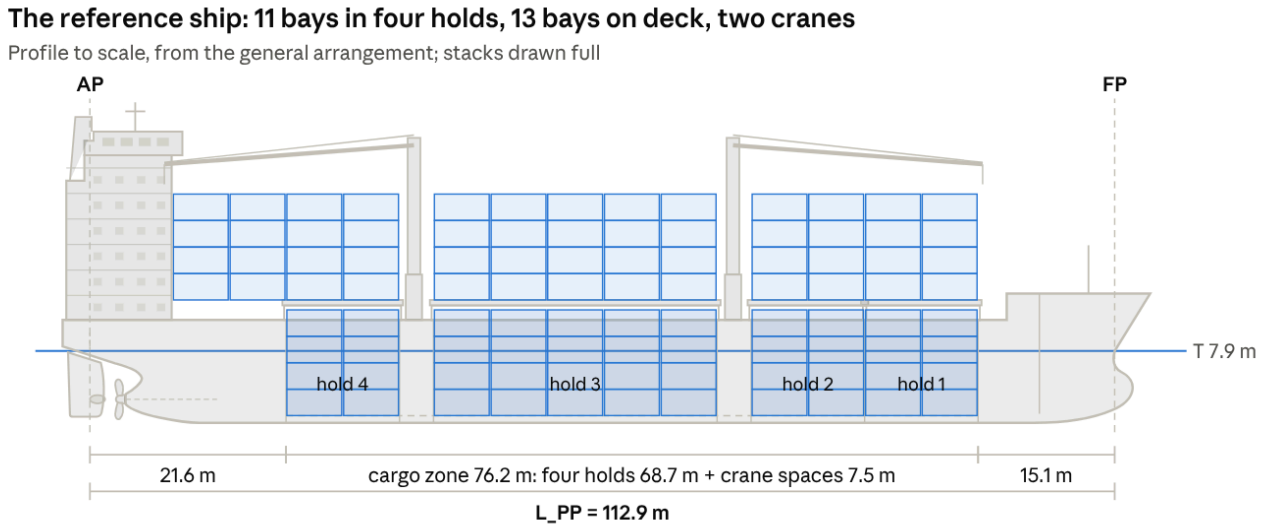


Figure I.10 · Reference ship, profile to scale

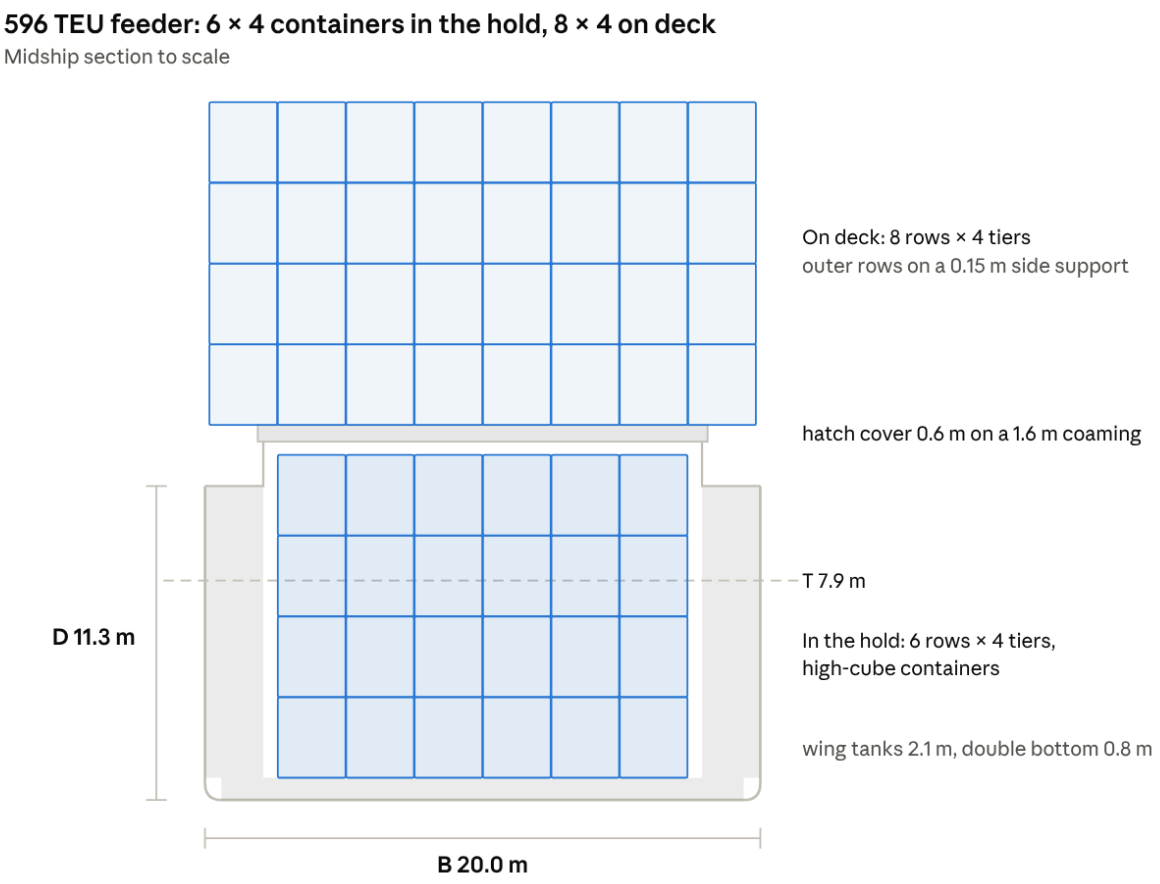


Figure I.11 · Reference ship, midship section to scale

What the method takes from the reference ship. The method itself is general, but some of its inputs are taken from the reference ship, because statistics do not give them. They shape every result of the worked example, and they would be different for another ship. They are:

Input taken from the reference ship	Value	What it sets
Double bottom, wing tanks, hatch coaming, hatch cover	0.8, 2.1, 1.6 and 0.6 m	the depth and the beam around the containers
Aft peak and engine room; fore peak	21.6 m; 15.1 m	the lengths added to the cargo zone
Container height	high cube, 2.896 m	the height of each tier, and so the depth
Extra deck bays over the engine room	2	the deck capacity
Eye height above the bridge deck	1.8 m, assumed	the view over the deck stacks
Hold utilization	0.86, measured	the number of bays, and so the length; the formula value, 0.74, is used alongside it
Block coefficient	0.68, assumed	the hold utilization by formula and the displacement

For your own design, replace each of these with the values of your own reference ship, or justify keeping them. Where the manual compares a result with the reference ship, it says so at that point.

Part II · The method

The method at a glance

A container ship is sized by whole containers. The owner's brief gives the number of containers, the speed and the deadweight; the method turns them into a hull whose main dimensions are built around the cargo.

First top-down, then bottom-up. Statistics of built ships give first targets for the beam and the depth (top-down). The method then fixes how many rows, tiers and bays the holds take, and builds each main dimension from the containers and the allowances around them (bottom-up): the beam from the rows, the guides and the double hull; the depth from the tiers, the double bottom and the margin under the hatch cover, less the coaming; the length from the bays, the engine room and the peaks.

Then the checks. A sequence of checks confirms that the ship carries its deck cargo within the sight line from the bridge, floats at an acceptable draught and is stable. Two things are not checked but closed by iteration inside each candidate: the weights against the displacement, and the hold utilization with the computed block coefficient. A candidate that fails a check is dropped: its fix, one more row or one more tier, is another candidate that the method already tries.

Where the method comes from. The sequence follows the design flow charts of Alvariño et al. (1997), drawn there on separate pages. Figure II.1 puts them on one page and numbers every calculation with the section of this manual where it is done. It also shows what goes into each chart, what comes out of it, and which part this manual covers: sizing, weights, capacity and stability. The technical evaluation that ends in a price comes after this manual.

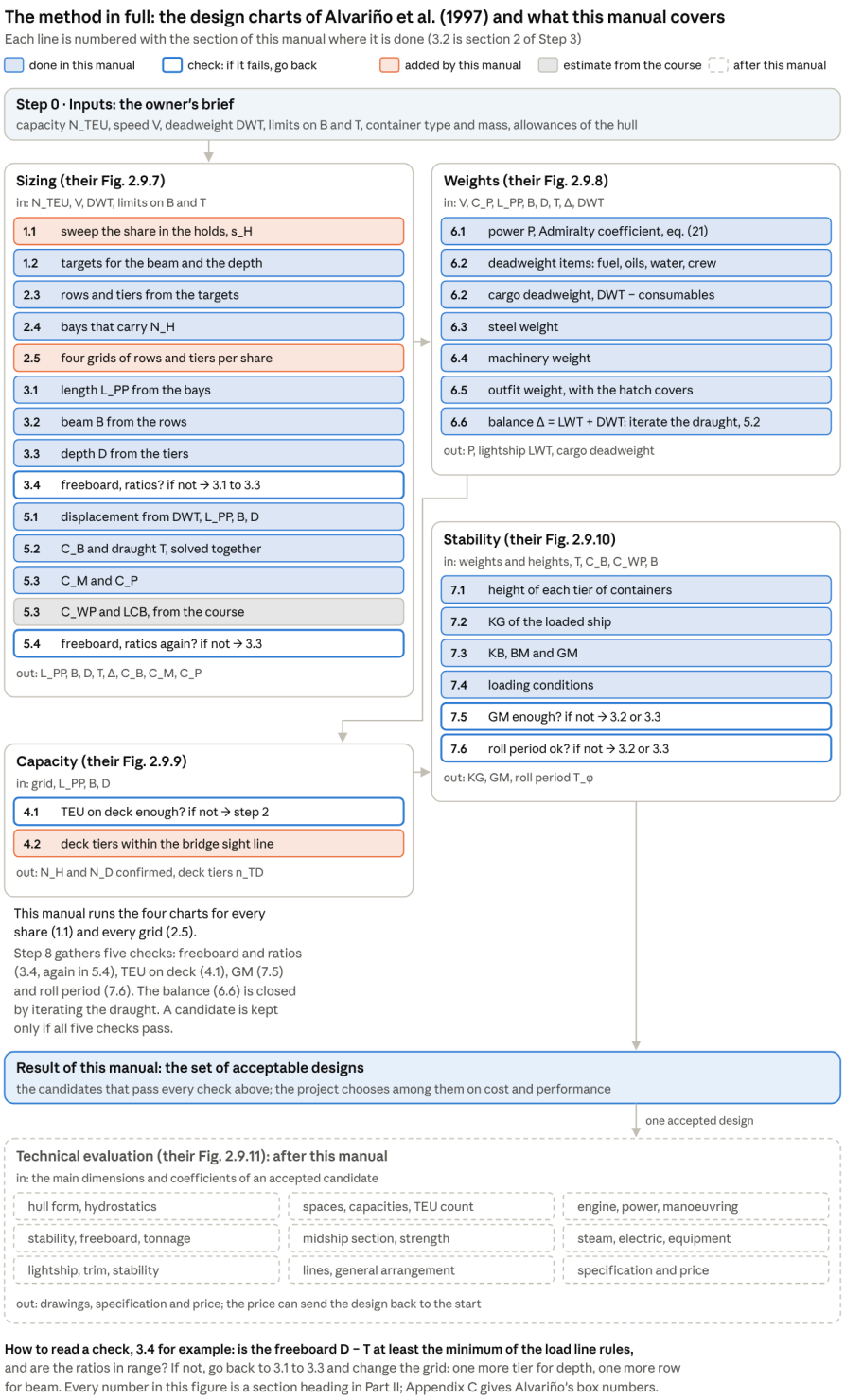


Figure II.1 · The method in full: the design charts of Alvariño et al. (1997), Figs. 2.9.7 to 2.9.11, numbered with the sections of this manual

The nine steps.

Step	What it decides	Main result	Equations
0 · Inputs	what the owner's brief asks and what the reference ship gives	`N_TEU`, `V`, `DWT`, limits on `B` and `T`, allowances	–
1 · Split and targets	the share of TEU in the holds, swept from 0.30 to 0.50, and the targets for each share	`N_H`, `N_D`, `B_0`, `D_0` per share	(1)–(5)

Step	What it decides	Main result	Equations
2 · Container grids	four grids per share: the rows and tiers that fit the targets, and one more row, one more tier, or both; bays for each	`n_R`, `n_T`, `n_B` per grid	(6)–(8)
3 · Dimensions	length, beam and depth of each grid, and whether their ratios are within the ranges of built ships	`L_PP`, `B`, `D`	(9)–(11)
4 · Deck and visibility	tiers on deck, within the sight line from the bridge	`n_TD`	(12)–(17)
5 · Displacement and form	the block coefficient and draught that float the ship	`Δ`, `C_B`, `T`, `C_M`, `C_P`	(18)–(20)
6 · Weights	lightship and deadweight in balance with `Δ`	`LWT`, `DWT`	(21)–(30)
7 · Stability	the height of the centre of gravity, the metacentric height and the roll period	`KG`, `GM`, `T_φ`	(31)–(36)
8 · Checks	keep the candidates that pass every check	the domain of candidates	–

Three ideas carry the method.

1. **Whole containers set the size.** The statistics of Step 1 · Split the TEU and set targets only give targets. The dimensions are rebuilt from whole rows, tiers and bays (Step 2 · Lay out the containers and Step 3 · Main dimensions from the container grid).
2. **The holds carry `N_H` by construction.** The number of bays is chosen so that the holds take the TEU assigned to them. Only the deck count must be checked against `N_D` (Step 4 · Capacity on deck and visibility).
3. **Iterate inside a candidate; drop it when a check fails.** Two loops run inside each candidate until they settle: the draught and `C_B`, until the hull floats its weights (5.2 Solve C_B and T and 6.6 Close the balance), and `k_U` with the computed `C_B`, which can change the number of bays and so the length (5.4 Close the iteration with step 2). A check is different. When the proportions, the freeboard or the stability fail, Alvariño et al. send the design back to change a count: one more tier for depth, one more row for beam. In this manual that change gives another grid of Step 2 · Lay out the containers, which the sweep already runs as a candidate of its own, so the failing candidate is simply dropped. If every candidate of a share fails, change the share or widen the bracket of Step 2 · Lay out the containers.

Where each variable comes from.

Variable	How it is obtained	Step	Reference ship
`N_TEU`	input: owner's brief	0	596
`s_H`	swept from 0.30 to 0.50 in steps of 0.05	1	0.38 real; 0.40 in Part III
`N_H`, `N_D`	`N_H = s_H · N_TEU`, `N_D` the rest	1	228 and 368 real; 238 and 358 at 0.40
`L_0`, `B_0`, `D_0`, `T_0`	regressions on `N_H`; only `B_0` and `D_0` feed Step 2 · Lay out the containers	1	117.9, 20.44, 9.97 and 7.58 m at 0.40
`n_R`, `n_T`	from `B_0` and `D_0`, rounded down; then one more of each	2	6 and 4 real
`k_U`	`0.5 C_B + 0.4`, or the value of a reference ship	2	0.74 from `C_B` 0.68; 0.86 real

Variable	How it is obtained	Step	Reference ship
`n_B`	the smallest even number that carries `N_H`	2	11 real; 12–18 for the four grids at 0.40
`L_PP`, `B`, `D`	built from the grid and the allowances	3	112.9, 20.0 and 11.3 m real
`n_TD`	deck capacity, then the sight line from the bridge	4	4 real
`C_B`, `T`	assumed in Step 2 · Lay out the containers; solved together in Step 5 · Displacement, C_B and draught	5	0.68 assumed, 7.9 m real

How each step is written. Every step opens with a card — purpose, inputs, outputs, equations, code — then gives the formulas with their source and a worked example (the Python functions that apply them are named in the card and listed in Appendix D), and closes with **Check before moving on**: the tests your numbers must pass before the next step.

Step 0 · Inputs

Step card	
Purpose	collect every number the method needs before sizing anything
Inputs	the owner's brief, the route study, one or more reference ships
Outputs	the input table below, filled in for your ship
Equations	none
Code	the *Inputs* block at the top of your program (Appendix D)

Alvariño et al. list the same starting data: the TEU split between holds and deck, the speed, the deadweight and any limits on beam or draught (§2.9.3.2). The allowances are defined, with typical values, in I.2.

Input	Typical source	Enters	Reference ship
Total capacity `N_TEU`	owner's brief	Step 1 · Split the TEU and set targets	596 TEU
Share in the holds `s_H`	swept over 0.30–0.50 (§2.9.3.1); a reference ship shows where in the range to look	Step 1 · Split the TEU and set targets	0.38
Container height: 8 ft 6 in or high cube	owner's brief	Step 2 · Lay out the containers	high cube
Service speed `V`	owner's brief	Step 1 · Split the TEU and set targets and Step 5 · Displacement, C_B and draught	16.5 kn

Input	Typical source	Enters	Reference ship
Deadweight `DWT` and design mass per TEU	owner's brief	Step 5 · Displacement, C_B and draught, Step 6 · Weights and power and Step 7 · Stability	8,893 t (reference ship). In the project the brief gives the cargo, `N_TEU` · 10 t, and a first `DWT` = cargo / 0.75 (Fig. A.49); 6.2 Deadweight items replaces it with the sum of the items
Limits on `B` and `T` (canals, ports)	route study	Step 2 · Lay out the containers, Step 3 · Main dimensions from the container grid and Step 5 · Displacement, C_B and draught	none
Aft peak and engine room `L_AP + L_ER`; fore peak `L_FP`	similar ships	Step 3 · Main dimensions from the container grid	21.6 m; 15.1 m
Double hull `w_DH`, double bottom `h_DB`, coaming `h_CO`, margin under the hatch cover `c_HC`, camber `c_DK`, hatch cover `t_cover`	a reference ship; otherwise the ranges of I.2	Step 2 · Lay out the containers, Step 3 · Main dimensions from the container grid and Step 4 · Capacity on deck and visibility	2.1, 0.8, 1.6, 0.49, 0 and 0.6 m
Guide clearances `g_LB`, `g_LF`, `g_BC`, `g_BF`	Table 2.9.2 (I.2)	Step 2 · Lay out the containers and Step 3 · Main dimensions from the container grid	lower values: 0.115, 0.180, 0.15 and 0.08 m
Extra deck bays `n_l` and deckhouse position	general arrangement of a reference ship	Step 4 · Capacity on deck and visibility	2 bays over the engine room; deckhouse aft
Assumed `C_B` for the first `k_U`	a reference ship	Step 2 · Lay out the containers	0.68
Endurance, crew	owner's brief	Step 6 · Weights and power	–

Check before moving on.

- Every row of the table has a value and a source.
- The container height is the one the brief asks for; if it is high cube, expect a deeper ship than the statistics give.
- If a reference ship's hold TEU and grid are known, its `k_U` is computed (I.3).

Step 1 · Split the TEU and set targets

Step card	
Purpose	split the capacity between holds and deck, and get first estimates of the dimensions
Inputs	`N_TEU`; the range of `s_H`
Outputs	for each share: `N_H`, `N_D`, `L_0`, `B_0`, `D_0`, `T_0`
Equations	(1) to (5)

Step card	
Code	`split`, `targets`, `length_from_trend`, `share_matching_trend`

Alvariño et al. call the split between holds and deck the main problem of the project, because the two parts are limited by different things: the hull for the holds, stability and visibility for the deck (§2.9.3.0; Figure I.3). Fix the split first, then take targets for the dimensions.

1.1 Split the TEU

Take the share in the holds from a reference ship. If there is no better data, use 30–50 % in the holds and 70–50 % on deck (§2.9.3.1), eq. (1):

$$N_H = s_H\, N_{TEU}, \qquad N_D = N_{TEU} - N_H, \qquad 0.30 \le s_H \le 0.50 \tag{1}$$

The brief does not fix `s_H`, so do not pick one value: sweep it in steps of 0.05 and run Step 1 · Split the TEU and set targets to Step 4 · Capacity on deck and visibility for each share. The range 0.30 to 0.50 is the one Alvariño et al. give for the ships behind their statistics (§2.9.3.1). For a ship of another size or service, take the range from the share in the holds of built ships like it. A reference ship's share tells you where in the range to expect the best candidates, not which one to keep.

1.2 Targets from the TEU in the holds

Alvariño et al. fit `L_PP`, `B`, `D` and `T` to the TEU in the holds of ships built with 8 ft 6 in containers (app. 2.9). Their `T` is the freeboard draught, the draught at the summer load line (I.4). For ships up to Panama beam, `B` ≤ 32.2 m, eqs. (2) and (3):

$$L_0 = 113.5 + 0.082\, N_H - \frac{3607.5}{N_H}, \qquad B_0 = \exp\left[0.674 - 2.7\cdot 10^{-4} N_H + 0.44 \ln N_H\right] \tag{2}$$

$$D_0 = \exp\left[0.249 - 5.36\cdot 10^{-6} N_H + 0.375 \ln N_H\right], \qquad T_0 = -6.05 + 2.49 \ln N_H \tag{3}$$

For post-Panamax ships, `B` > 32.2 m, eqs. (4) and (5):

$$L_0 = 279 + 0.0183\, N_H - \frac{1.19\cdot 10^5}{N_H}, \qquad B_0 = 42.8 - 5.72\cdot 10^{-3} N_H + 1.69\cdot 10^{-6} N_H^2 \tag{4}$$

$$D_0 = 18.8 + 2.11\cdot 10^{-3} N_H, \qquad T_0 = 11.6 + \frac{1.73\cdot 10^4}{N_H} - \frac{3.03\cdot 10^7}{N_H^2} \tag{5}$$

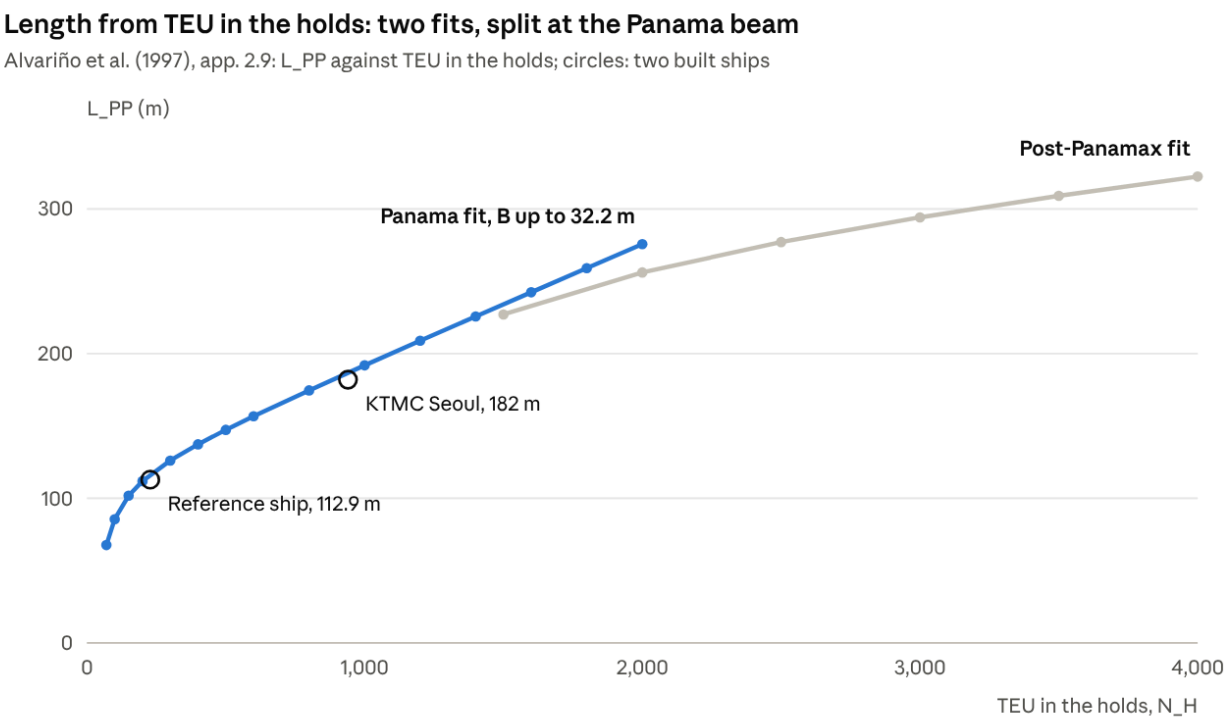


Figure II.2 · Alvariño et al. (1997), app. 2.9: Panama fit 70–2,000 TEU in the holds, post-Panamax fit 1,500–4,000

Ships with 1,000–1,500 TEU in the holds are usually Panama size; above about 2,000 they are all post-Panamax (app. 2.9.A1). Only `B_0` and `D_0` are used in Step 2 · Lay out the containers; `L_0` and `T_0` are kept for comparison, and `T_0` for the ratio check of Step 3 · Main dimensions from the container grid.

1.3 Cross-check with built ships

Appendix A of Papanikolaou plots `L_PP` against the total TEU (Fig. A.47) and `B` and `T` against `L_PP` (Fig. A.48), from the statistics of Kristensen (2013). Use it to check that the targets of 1.2 Targets from the TEU in the holds are plausible for the total capacity. Two built ships are shown: the reference ship, 3 m below the trend, and KMTC Seoul, a 2,540 TEU ship of 182 m used again in III.2, where the trend gives about 205 m.

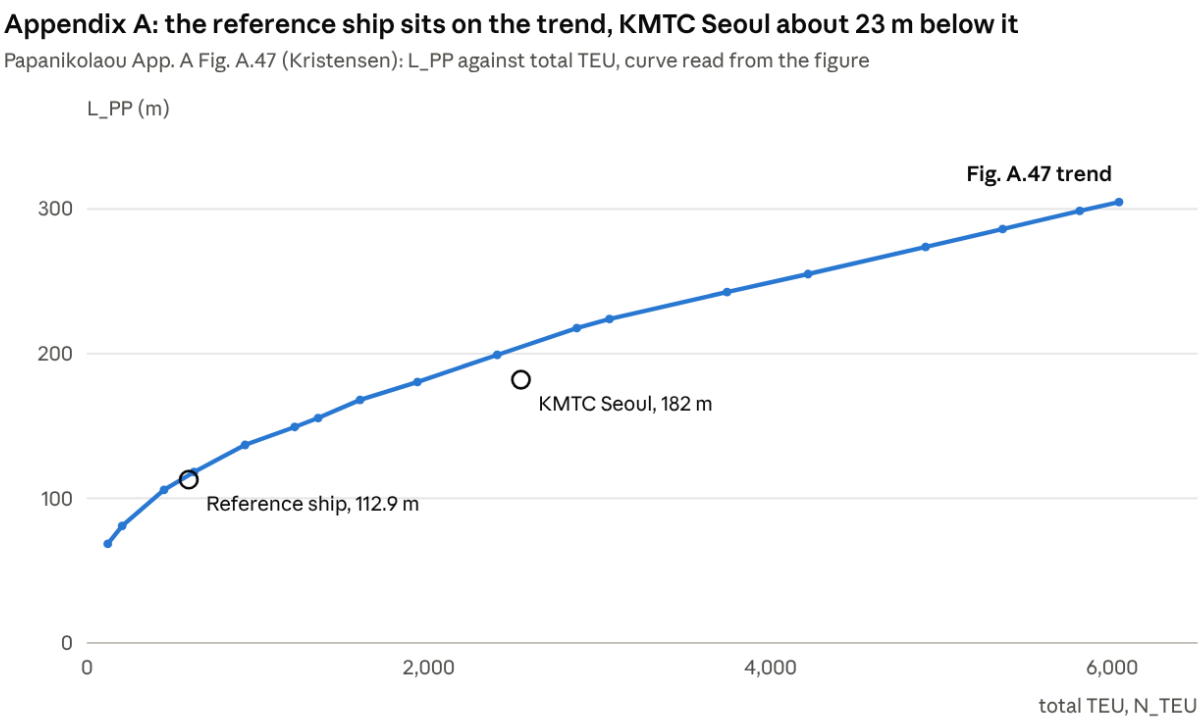


Figure II.3 · Papanikolaou App. A Fig. A.47 (Kristensen): trend line read from the figure, 60–305 m

An optional use: which share fits the total? Each share `s_H` gives a different `N_H`, so eq. (2) gives a different `L_0`; Appendix A gives one length for the total TEU. The share where the two agree is a good centre for the sweep when you want the design to align with built ships. It is not a required step and not a constraint, because real ships scatter around the trend. For the reference ship, `L_0` runs from 108.0 m at 0.30 to 125.8 m at 0.50 and meets the Appendix A length, 116.0 m, at `s_H` = 0.38; the real ship carries 38 % of its TEU in the holds.

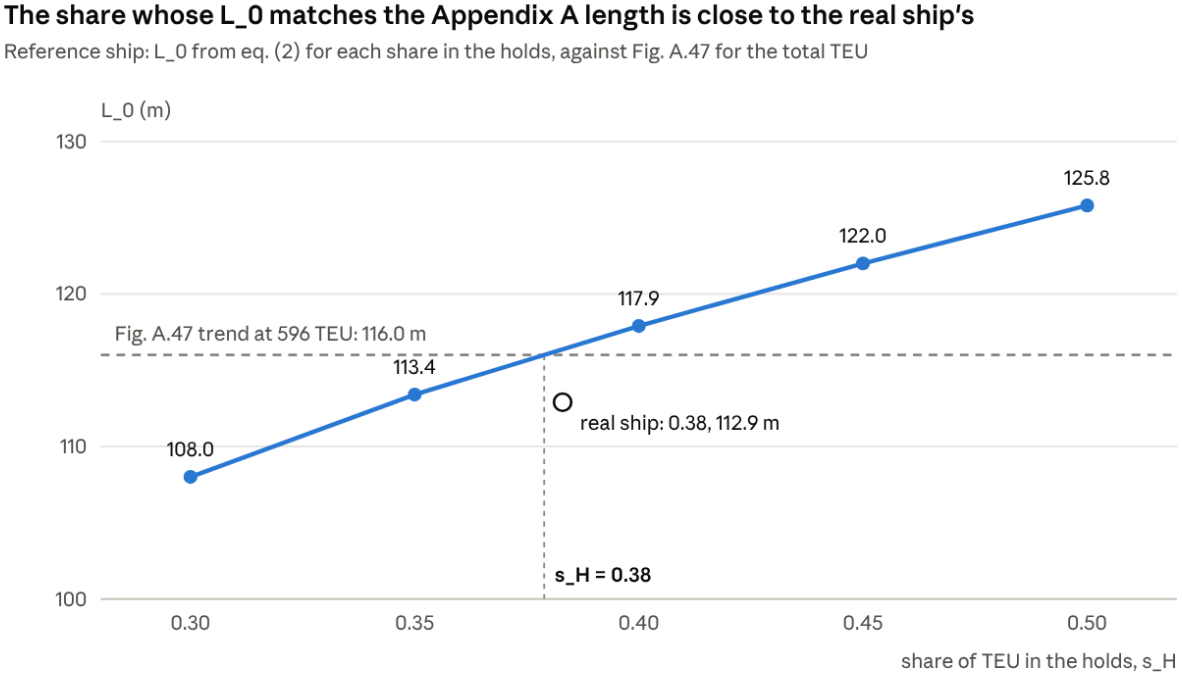


Figure II.4 · Reference ship: eq. (2) for the five shares of the sweep, against Fig. A.47

Check before moving on.

- `N_H + N_D = N_TEU` for every share.
- The right fit is used: eqs. (2) and (3) up to Panama beam, (4) and (5) above it, and `N_H` is inside the range of the fit.
- `L_0` has been compared with the Appendix A length for the total TEU, and any large difference is explained.
- For high cubes, `D_0` is expected to be low: Step 2 · Lay out the containers corrects it with one more tier.

Step 2 · Lay out the containers

Step card	
Purpose	turn the targets into three whole numbers: rows, tiers and bays
Inputs	`B_0`, `D_0`, `N_H` from Step 1 · Split the TEU and set targets; the allowances of I.2; `k_U` from I.3
Outputs	four grids per share, each with `n_R`, `n_T`, `n_B`
Equations	(6) to (8)
Code	`rows`, `tiers`, `hold_utilization`, `bays`

A ship cannot carry 6.39 rows or 3.53 tiers. Step 2 · Lay out the containers rounds the targets of Step 1 · Split the TEU and set targets to whole containers; Step 3 · Main dimensions from the container grid then builds the dimensions back from these whole numbers, so that every metre of beam and depth is explained by a container, a guide or a structure. Alvariño et al. give this order explicitly: once the share in the holds is fixed, rows and tiers follow from the beam and depth, and bays from the TEU in the holds (§2.9.3.2).

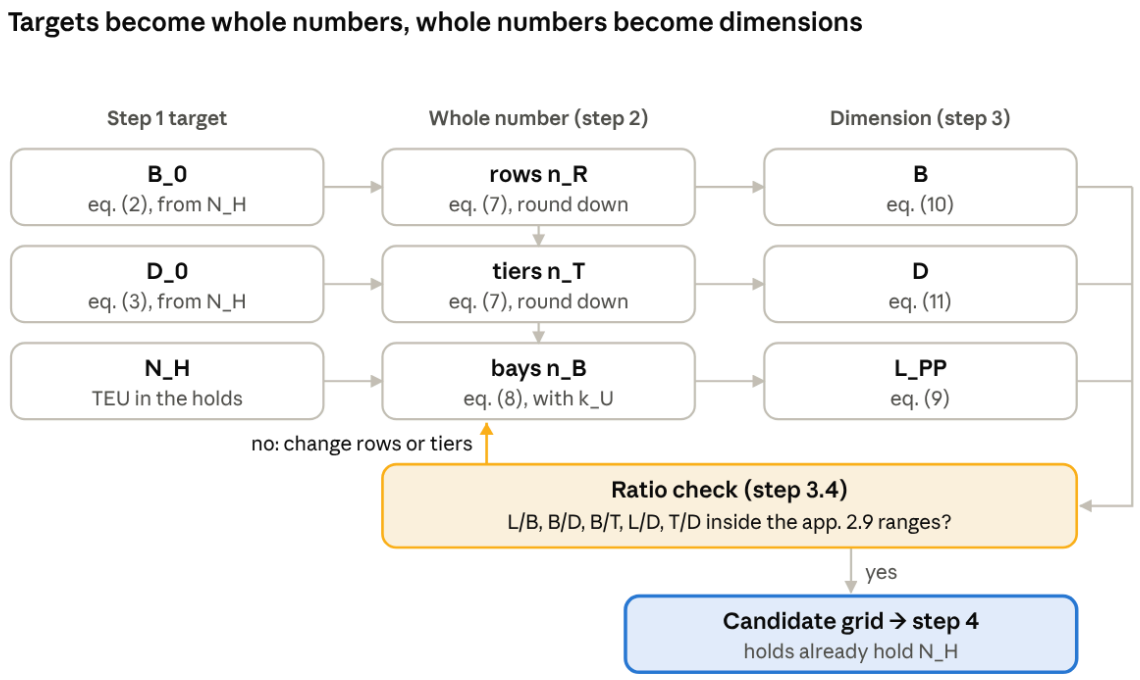


Figure II.5 · From step 1 to step 3: each target becomes a whole number, each whole number a dimension, and the ratios are checked

2.1 What you need from Part I

The container sizes and pitches are in I.1 and I.2 (Figures I.1, I.4), the allowances `w_DH`, `h_DB`, `h_CO`, `c_HC`, `g_BC`, `g_BF` in the tables of I.2, and the hold utilization `k_U` in I.3. Use the values of the reference ship when you have them (I.5); otherwise the ranges of Alvariño et al. Each tier takes the container height plus a 13 mm gap (§3.8.6.1): 2.604 m for 8 ft 6 in, 2.909 m for high cubes. Each row takes 2.54 m, the container width plus 100 mm.

2.2 One hold is one 40 ft bay

A hold takes two 20 ft containers end to end, or one 40 ft container, between two transverse bulkheads. Its length is the containers plus the guide clearances `g_LB` and `g_LF` (I.2), eq. (6), from Eq. 2.9.1:

$$l_{hold} = 2\,g_{LB} + 2\,l_c + g_{LF} \approx 12.53\text{–}12.56\ \text{m} \qquad (6)$$

This is why the number of bays `n_B` is always even: two 20 ft bays per hold. In a first estimate, §3.8.6.1 uses 6.55 m per 20 ft bay, which includes the gap between 40 ft containers and a share of the bulkheads (Step 3 · Main dimensions from the container grid, 3.1 Length).

2.3 Rows and tiers

Rows come from the beam target, tiers from the depth target, both rounded down, eq. (7), from Eqs. 3.8.30 and 3.8.31:

$$n_R = \left\lfloor \frac{B_0 - 2\,w_{DH}}{2.54} \right\rfloor, \qquad n_T = \left\lfloor \frac{D_0 + c_{DK} + h_{CO} - h_{DB} - c_{HC}}{h_c + 0.013} \right\rfloor \qquad (7)$$

Read the tier formula as a height budget (Figure I.5): from the top of the double bottom to the underside of the hatch cover there is `D + c_DK + h_CO − h_DB − c_HC` of free height, and each tier takes `h_c + 0.013`.

2.4 Bays

The full grid `n_B · n_R · n_T` is reduced by the hold utilization `k_U`, because the hull narrows at the ends and the corner cells are lost (I.3, Figure I.6). Take the smallest even `n_B` that satisfies eq. (8), from Eq. 2.9.4:

$$n_B\,n_R\,n_T\,k_U \geq N_H, \qquad k_U = 0.5\,C_B + 0.4 \qquad (8)$$

`C_B` is not known until Step 5 · Displacement, C_B and draught, so start with the `C_B` of a reference ship and repeat once `C_B` is computed; Alvariño et al. say the same (§2.9.3.2). `k_U` is not swept like `s_H` or the grid, because it follows from the hull of each candidate; it is iterated with `C_B` (Step 5 · Displacement, C_B and draught, 5.4 Close the iteration with step 2), and its uncertainty, 0.70 to 0.86 (I.3), is shown as a sensitivity in III.1.

2.5 Bracket the targets: four grids per share

The targets `B_0` and `D_0` are averages of built ships, and they almost never fall on a whole number of rows or tiers. Eq. (7) rounds down, which gives the largest grid that stays inside the targets. The next whole number is just as close to the statistics, and with high cubes, whose depth target comes from ships with 8 ft 6 in containers, it is often the better ship.

So the method does not choose: it keeps the two whole numbers on either side of each target, `n_R` or `n_R + 1` rows and `n_T` or `n_T + 1` tiers, and lets the checks of Step 3 · Main dimensions from the container grid to Step 8 · Check and iterate decide. Two choices for rows times two for tiers give four grids. The rule does not depend on the size of the ship: it only turns a point of the statistics into the whole numbers around it.

Grid	Rows	Tiers	Effect on the ship
1	`n_R` of eq. (7)	`n_T` of eq. (7)	narrowest and shallowest, so the longest
2	`n_R`	`n_T + 1`	one tier deeper, fewer bays
3	`n_R + 1`	`n_T`	one row wider, fewer bays
4	`n_R + 1`	`n_T + 1`	widest and deepest, so the shortest

Going further than plus one would leave the bracket: the ship would be more than a whole container away from both targets, so it is not done. **When the beam is fixed** by a canal, a port or the outreach of the cranes, take `n_R` from that beam instead of from `B_0` and bracket only the tiers: two grids per share. Each grid gets its own `n_B` from eq. (8); Step 3 · Main dimensions from the container grid builds its dimensions and removes those that are out of proportion.

Example: the four grids at `s_H` = 0.40. For the reference ship brief, Step 1 · Split the TEU and set targets gives `N_H` = 238, `B_0` = 20.44 m and `D_0` = 9.97 m. With `w_DH` = 2.1 m, `h_CO` = 1.6 m, `h_DB` = 0.8 m, `c_DK` = 0, high cubes and `c_HC` = 0.49 m, which is 4 % of 12.19 m, the shorter side of a hatch cover over a 40 ft bay (Eq. 2.9.3 gives 4–5 % of the shorter side):

1. Rows by eq. (7): (20.44 − 2 × 2.1) / 2.54 = 6.39. Rounded down, 6 rows; the beam target would hold 0.39 of a row more.
2. Tiers by eq. (7): (9.97 + 1.6 − 0.8 − 0.49) / 2.909 = 3.53. Rounded down, 3 tiers; 0.53 of a tier is lost.
3. The four grids are 6 × 3, 6 × 4, 7 × 3 and 7 × 4.
4. Bays by eq. (8) with `k_U` = 0.74 (`C_B` = 0.68): for grid 1, 238 / (6 × 3 × 0.74) = 17.9, so 18 bays; for grid 4, 238 / (7 × 4 × 0.74) = 11.5, so 12 bays.

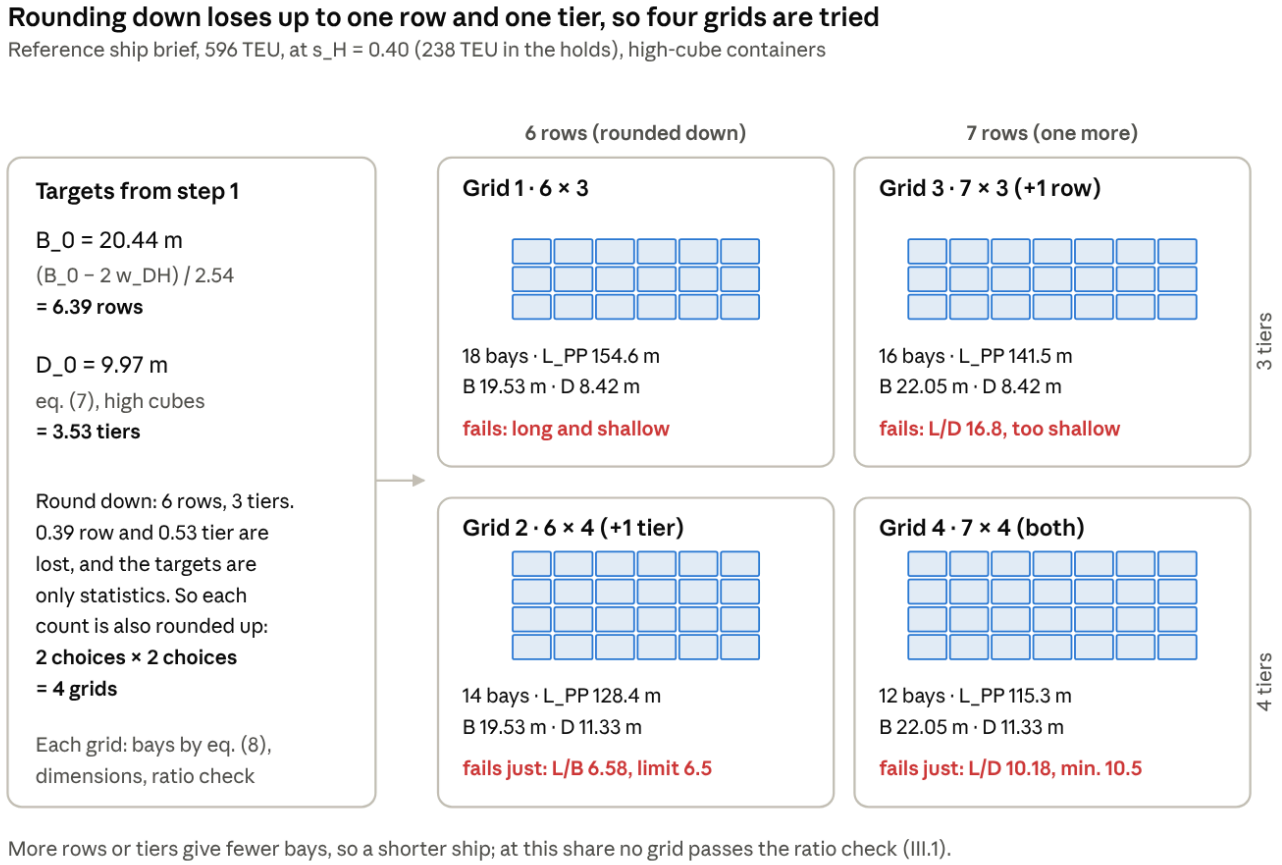


Figure II.6 · The four grids of one share: reference ship brief at s_H = 0.40

No grid of this share passes every ratio of Step 3 · Main dimensions from the container grid, 3.4 Check the ratios. Grids 1 and 3 have only three tiers and come out too long and shallow; grid 2 is just over the `L/B` limit, and grid 4 just under the `L/D` limit. One more row or tier makes each bay hold more, so fewer bays are needed and the ship is shorter. The other shares of the sweep give the grids that pass (III.1).

Check before moving on.

- `n_R` and `n_T` are rounded down in eq. (7), and the four grids add 0 or 1 to each.
- `n_B` is even, and `n_B · n_R · n_T · k_U` is at least `N_H`.
- The `C_B` used for `k_U` is written down, so that Step 5 · Displacement, C_B and draught can show whether it must be repeated.
- The tier height matches the container: 2.604 m for 8 ft 6 in, 2.909 m for high cubes.

Step 3 · Main dimensions from the container grid

Step card	
Purpose	build `L_PP`, `B` and `D` of each grid, then keep only the grids in proportion
Inputs	`n_R`, `n_T`, `n_B` of each grid (Step 2 · Lay out the containers); `L_AP`, `L_ER`, `L_FP`, the allowances (I.2); `T_0` (Step 1 · Split the TEU and set targets)
Outputs	`L_PP`, `B`, `D` and a pass or fail for each grid
Equations	(9) to (11), ratio table of 3.4 Check the ratios
Code	`length`, `beam`, `depth`, `ratio_limits`, `check_ratios`

The dimensions are now built from the inside out: containers, then clearances, then structure. Unlike the targets of Step 1 · Split the TEU and set targets, each dimension here can be traced to a part of the ship.

3.1 Length

The length is the cargo zone plus three parts that carry no containers in the holds: the aft peak `L_AP`, the engine room `L_ER` and the fore peak `L_FP`. Take them from similar ships (§2.9.3.1); for the reference ship, `L_AP + L_ER` = 21.6 m and `L_FP` = 15.1 m (I.5). With `t_BH` the thickness of a transverse bulkhead and `n_HO` the number of holds, eq. (9), from Eq. 2.9.1:

$$L_{PP} = L_{AP} + L_{ER} + L_{FP} + L_C, \qquad L_C = (2\,g_{LB} + 2\,l_c + g_{LF})\,n_{HO} + t_{BH}\,(n_{HO} - 1)\,, \qquad n_{HO} = n_B/2 \tag{9}$$

In a first estimate, §3.8.6.1 uses 6.55 m per 20 ft bay for the whole cargo zone (Eq. 3.8.32), so `L_C ≈ 6.55 n_B`. The code of Appendix D uses this form. Each extra pair of bays adds 13.1 m to the ship, which is why the length moves in steps of 13.1 m in all the tables of this manual.

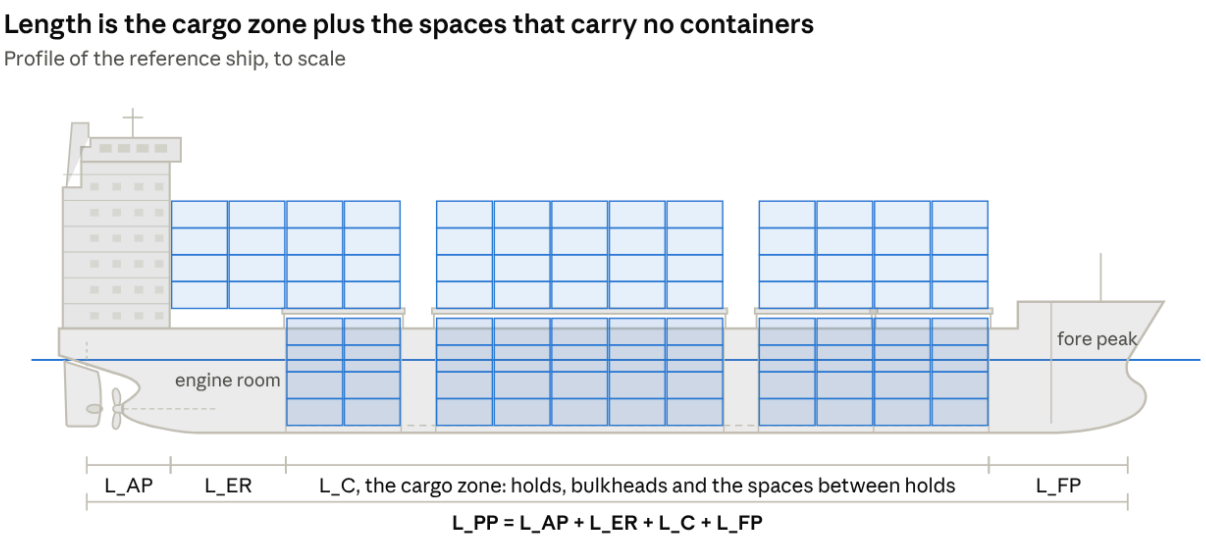


Figure II.7 · Profile: how the length adds up from holds, engine room and peaks

3.2 Beam

Rows, guides and the double hull on both sides, eq. (10), from Eq. 2.9.2 (Figure I.4):

$$B = 2\,w_{DH} + 2\,g_{BC} + n_R\,b_c + g_{BF}\,(n_R - 1) \qquad (10)$$

If the ship must pass a canal, check `B` against the lock: 32.2 m for the original Panama locks (app. 2.9), 51.25 m for the New Panamax locks. For a feeder this is not binding, but the same check matters for a larger brief.

3.3 Depth

The tiers in the hold must fit under the hatch cover. Eq. (11) is the tier formula of eq. (7), Eq. 3.8.31, solved for `D`:

$$D = n_T\,(h_c + 0.013) + h_{DB} + c_{HC} - h_{CO} - c_{DK} \qquad (11)$$

Eq. 2.9.3 of the same text is the simpler form, without the 13 mm gaps and with no camber (`c_DK` = 0). Note what eq. (11) says: a higher coaming `h_CO` gives a lower depth for the same tiers, because part of the top tier then stands inside the coaming, above the deck.

Rows set the beam, tiers set the depth

Midship section, schematic, not to scale

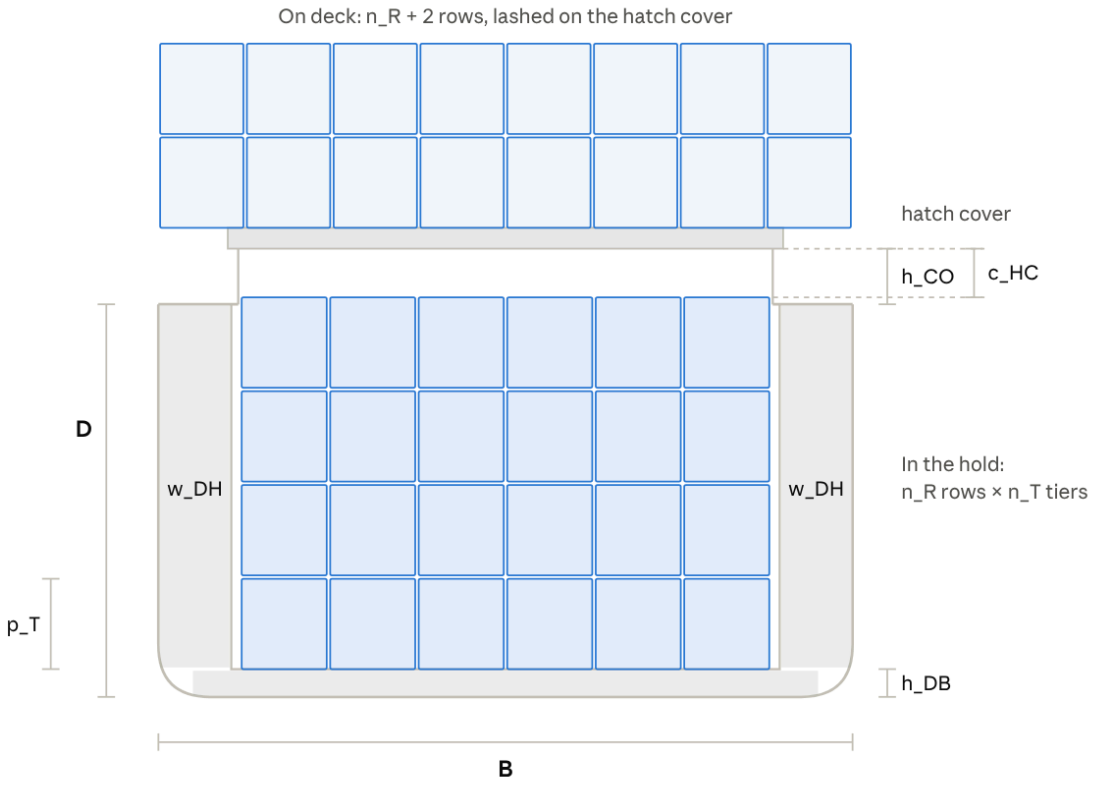


Figure II.8 · Midship section: how B and D follow from rows and tiers; p_T is the tier pitch (schematic)

3.4 Check the ratios

Compare the ratios with the ranges of app. 2.9, for the row that matches `N_H`:

Ships	N_H	L/B	B/D	B/T	L/D	T/D	Fn
Panama	< 1,000	5.2–6.5	1.7–2.2	2.4–3.2	10.5–13.0	0.63–0.79	0.23–0.26
Panama	1,000–1,500	5.7–7.5	1.6–1.9	2.4–3.1	10.5–12.7	0.57–0.70	0.23–0.26
Panama	1,500–2,000	6.5–8.5	1.5–1.7	2.4–2.9	11.0–13.5	0.55–0.65	0.23–0.25
Post-Panamax	1,500–3,000	6.4–7.3	1.6–1.75	2.8–3.5	10.5–12.2	0.5–0.6	0.22–0.25
Post-Panamax	> 3,000	6.7–7.5	1.6–1.75	2.7–3.3	11.2–12.5	0.5–0.6	0.21–0.24

Use `T_0` for `B/T` and `T/D`, the freeboard draught of eq. (3) or (5). Alvariño et al. do not state which draught their `B/T` and `T/D` ranges use; their draught regression uses the freeboard draught.

`T_0` is not a limit. The draught that Step 5 · Displacement, C_B and draught and Step 6 · Weights and power compute is checked against the draught limit of the brief (ports, canals; Step 0 · Inputs) and against the freeboard, `D − T` at least the minimum freeboard. `Fn` is checked in Step 5 · Displacement, C_B and draught, once the speed and length are known.

Freeboard. Check here too that the depth leaves at least the minimum freeboard of the load line rules above the target draught, `D − T_0`. Section 5.4 Close the iteration with step 2 repeats this check, and the ratios, with the draught that floats the ship. A grid that fails is not repaired, it is dropped, because the other grids of Step 2 · Lay out the containers already cover one more row or one more tier.

Example: the four grids at `s_H` = 0.40. With `T_0` = 7.58 m, `N_H` = 238, so the first row of the table applies:

Grid	n_B	L_PP (m)	B (m)	D (m)	L/B	B/D	B/T	L/D	T/D	Result
1 · 6 × 3	18	154.6	19.53	8.42	**7.92**	**2.32**	2.58	**18.37**	**0.90**	fails
2 · 6 × 4	14	128.4	19.53	11.33	**6.58**	1.72	2.58	11.34	0.67	fails, just
3 · 7 × 3	16	141.5	22.05	8.42	6.42	**2.62**	2.91	**16.81**	**0.90**	fails
4 · 7 × 4	12	115.3	22.05	11.33	5.23	1.95	2.91	**10.18**	0.67	fails, just

Values in bold are out of range. The three-tier grids fail on depth: the ship is too long for how shallow it is. The worked example (III.1) calls grids 1, 2 and 4 candidates A, B and C.

Check before moving on.

- `L_PP` changes in steps of 13.1 m (two bays) between grids; any other step means an error in `n_B`.
- `B` and `D` can be traced term by term to Figure II.8.
- The ratio row matches `N_H`, and `B/T`, `T/D` use `T_0`, not a design draught.
- At least one grid of the sweep passes; if none does for any share, revisit the allowances of I.2 before changing the method.

Step 4 · Capacity on deck and visibility

Step card	
Purpose	find the deck tiers that carry `N_D`, and check them against the sight line from the bridge
Inputs	`N_D` (Step 1 · Split the TEU and set targets); `n_B`, `n_R`, `L_PP`, `D` of each grid (Step 2 · Lay out the containers and Step 3 · Main dimensions from the container grid); `n_l`, `h_CO`, `t_cover`, `x_eye` (Step 0 · Inputs)
Outputs	`n_TD`; the bridge and eye height; the tiers allowed at each bay
Equations	(12) to (17)
Code	`deck_teu`, `deck_tiers_needed`, `bridge_height`, `tiers_visible`

The holds already carry `N_H` by construction (Step 2 · Lay out the containers). This step finds the deck tiers that carry `N_D`, checks them against the sight line from the bridge, and confirms the TEU in the holds, on deck and in total. Recall from I.1 why the deck matters: on a feeder, more than half of the TEU are on deck.

4.1 Containers on deck

On deck the stacks stand on the hatch covers and reach two more rows than in the hold, because the deck is not narrowed by the double hull (Figure II.8), eq. (12), from Eq. 2.9.5:

$$N_D \le (n_B + n_l)\,(n_R + 2)\,n_{TD}, \qquad n_l = 0, 1, 2, \qquad n_{TD} \le 5 \qquad (12)$$

`n_l` counts the extra deck bays that the position of the deckhouse allows, for example over the engine room; the reference ship has 2 (I.5). Take the smallest `n_TD` that carries `N_D`. If five tiers are not enough, the split of Step 1 · Split the TEU and set targets puts too much on deck: raise `s_H` and repeat.

4.2 Visibility

SOLAS V/22 limits the blind sector ahead of the bow: from the conning position (the place on the bridge from which the ship is navigated), the sea surface must be visible from no more than two ship lengths, or 500 m if that is less, ahead of the bow, for every draught, trim and deck load. Draw a straight line from the eye to the sea at that distance; the top of each deck stack must stay below it, eqs. (13) and (14):

$$z(x) = T + (z_{eye} - T)\,\frac{x_{sea} - x}{x_{sea} - x_{eye}}, \qquad x_{sea} = x_{bow} + \min\left(2\,L_{OA},\ 500\ \text{m}\right) \tag{13}$$

$$n_{TD}(x) = \left\lfloor \frac{z(x_{fwd}) - z_{base}}{h_c + 0.013} \right\rfloor, \qquad z_{base} = D + h_{CO} + t_{cover} \tag{14}$$

Here `x` is measured forward from the aft perpendicular and `z` up from the base line (Figure I.5); `x_eye` and `z_eye` are the position and height of the eye on the bridge, `x_bow` the forward end of the ship, `L_OA` the length overall, `x_fwd` the forward end of the bay and `t_cover` the hatch-cover thickness. Alvariño et al. give the height of the navigation bridge above base for ships with the engine room aft (Eq. 3.8.36); this manual adds 1.8 m for the eye above the bridge deck, eq. (15):

$$z_{BR} = 0.22\,L_{PP} + 0.28\,D^{1.56} - 0.02\,L_{PP}^{0.806}\,D^{1.1}, \qquad z_{eye} = z_{BR} + 1.8\ \text{m} \tag{15}$$

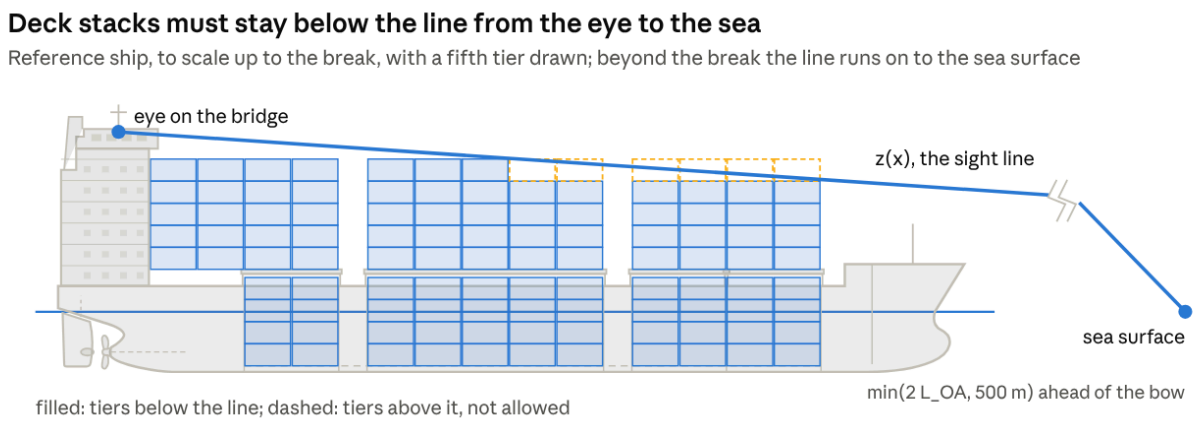


Figure II.9 · Bridge visibility: the sight line caps the deck stacks, more at the bow than amidships

The line falls towards the bow, so the forward bays take fewer tiers than those near the bridge. The program of Alvariño et al. counts deck TEU the same way, with the line at 2 `L_PP` from the forward perpendicular and at most five tiers on the hatch covers (§3.8.6.2); this manual uses the SOLAS distance. Two simplifications are made: the draught is `T_0` (a lighter, trimmed ship sees worse), and `L_OA` is taken as `L_PP` (the code of Appendix D passes `L_PP` for both). The stability check of Step 7 · Stability can lower `n_TD` further.

Example: grid 4 at `s_H` = 0.45, the first share where a grid passes Step 3 · Main dimensions from the container grid, 3.4 Check the ratios (III.1). `n_B` = 14, `n_R` = 7, `N_D` = 328, `n_l` = 2.

1. Deck by eq. (12): (14 + 2) · (7 + 2) · `n_TD` ≥ 328. Two tiers give 288, three give 432: `n_TD` = 3.
2. Bridge by eq. (15), with `L_PP` = 128.4 m and `D` = 11.33 m: `z_BR` = 26.1 m, `z_eye` = 27.9 m above base, at `x_eye` = 10.8 m (the deckhouse over the engine room).
3. The first deck tier stands at `z_base` = 11.33 + 1.6 + 0.6 = 13.53 m. The sight line meets the sea 256.8 m ahead of the bow, at `T_0` = 7.87 m.
4. Tiers under the line, eq. (14), at the forward end of each bay:

Bays (from aft)	Forward end `x_fwd`	Tiers allowed
1–6	28.2–60.9 m	4
7–14	67.5–113.3 m	3

Every bay can carry the three tiers needed, so no TEU are lost: the deck carries 432 TEU, 104 more than `N_D`. The sight line limits the forward bays to three tiers, the number this candidate needs.

4.3 Second estimates

Two regressions of §3.8.6 check the counts. For the holds, with `p = n_R · n_T` at midship, eq. (16), from Eqs. 3.8.28 and 3.8.29:

$$N_H = 15.67\, p^{0.6589}\, n_B^{0.5503}\, C_B^{0.598} - 126 \quad (L_{PP} < 185\ \mathrm{m}), \qquad N_H = 0.011\, p^{1.746}\, n_B^{1.555}\, C_B^{3.505} + 704 \quad (L_{PP} > 185\ \mathrm{m}) \tag{16}$$

For the deck, ships with the engine room aft, with `P` the maximum continuous rating of the main engine in brake horsepower (BHP), eq. (17), from Eq. 3.8.37:

$$N_D = 145\, L_{PP}^{0.36}\, B^{0.18} + 0.0032\, P^{1.18} - 1074 \tag{17}$$

The two forms of eq. (16) jump by about 24 % across 185 m, and eq. (17) can go negative for small ships (Appendix B). Use them as a second opinion only. Once the hull lines are drawn, the containers can be counted directly in the shape of the hull, which replaces all of these estimates.

Check before moving on.

- `n_TD` is at most 5; if not, raise `s_H`.
- The tiers allowed at the foremost bay are known, and the TEU lost forward are subtracted from eq. (12).
- The draught used for the sight line is stated; a ballast or trimmed condition sees worse.
- Eqs. (16) and (17) are used as a second opinion only, never to replace eqs. (8) and (12).

Step 5 · Displacement, C_B and draught

Step card	
Purpose	find the displacement, the block coefficient and the draught of each candidate
Inputs	`L_PP`, `B`, `D` (Step 3 · Main dimensions from the container grid); `DWT` and `V` from the brief; `LWT_1` of a similar ship
Outputs	`Δ`, `C_B`, `T`, `Fn`, `C_M`, `C_P`; a revised `n_B` if `k_U` changes
Equations	(18) to (20)
Code	`cb_katsoulis`, `solve_cb_t`, `froude_number`, `midship_coefficient`

With `L_PP` and `B` fixed by the grid, the draught and the block coefficient are the two unknowns left. Alvariño et al. solve them together (their Fig. 2.9.7). Note the change of logic from Step 1 · Split the TEU and set targets to Step 4 · Capacity on deck and visibility: there the hull was built around the cargo; here the hull must float the weights.

5.1 First displacement

Displacement is deadweight plus lightship (Eq. 2.9.10). For the first design, take the lightship of a similar ship of the same TEU class. For each later candidate `i`, scale it from that first design, marked with the subscript 1, eq. (18), from Eqs. 2.9.10 and 2.9.17:

$$\Delta_1 = DWT + LWT_1, \qquad LWT_i = LWT_1 \left(\frac{L_i}{L_1}\right)^{1.5} \frac{B_i}{B_1} \left(\frac{D_i}{D_1}\right)^{0.5} \tag{18}$$

Step 6 · Weights and power replaces `LWT` by an estimate built from its parts. Eq. (18) says the lightship grows fastest with length: a candidate two bays longer (13.1 m on 115 m) carries about 18 % more lightship.

When no similar ship gives a lightship, start from any draught: 6.6 Close the balance builds `LWT` from its parts and finds the draught. The worked example starts from `B` / 2.9.

5.2 Solve `C_B` and `T`

Two equations, two unknowns. The Katsoulis formula, with `L_PP`, `B` and `T` in metres and the service speed `V` in knots (Eq. 2.9.11), and the displacement in tonnes (Eq. 2.9.12), together eq. (19):

$$C_B = 0.8217\, L_{PP}^{0.42}\, B^{-0.3072}\, T^{0.1721}\, V^{-0.6135}, \qquad \Delta = 1.030\, C_B\, L_{PP}\, B\, T \tag{19}$$

The factor 1.030 instead of 1.025 covers the appendages and the shell plating (§2.9.3.2). Start from any draught and repeat until `T` changes by less than 1 cm:

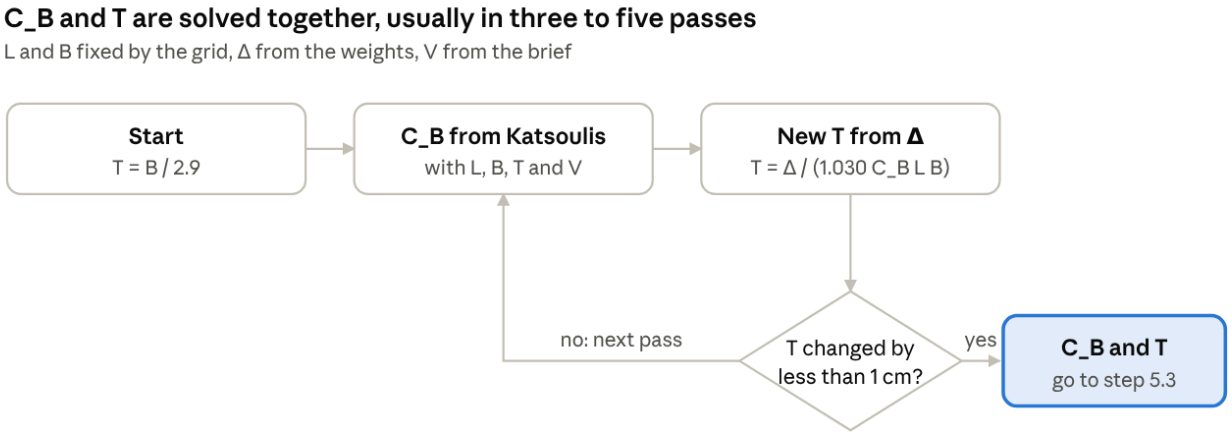


Figure II.10 · Eq. (19) as a loop: Katsoulis and the displacement solved together

How the loop converges, for the reference ship at 17 kn (`L_PP` = 112.9 m, `B` = 20.0 m, `Δ` = 12,494 t):

Pass	`T` in	`C_B`	`T` out
1	6.897 m (`B` / 2.9)	0.5843	9.194 m
2	9.194 m	0.6139	8.750 m
3	8.750 m	0.6087	8.825 m
4	8.825 m	0.6096	8.812 m
5	8.812 m	0.6094	8.815 m, change below 1 cm

The exponent of `V` is negative, so a lower speed gives a fuller hull: use the service speed of the brief and state it.

Example: the reference ship. Its service speed is 16.5 kn and its summer deadweight 8,893 t (I.5); its lightship is not published, so Step 5 · Displacement, C_B and draught is shown as a sensitivity. Take the displacement for the assumed `C_B` = 0.68 at `T` = 7.9 m, 1.030 × 0.68 × 112.9 × 20.0 × 7.9 = 12,494 t, which implies a lightship of 12,494 − 8,893 = 3,601 t, and solve eq. (19) for that displacement at the ship's speed and at speeds around it:

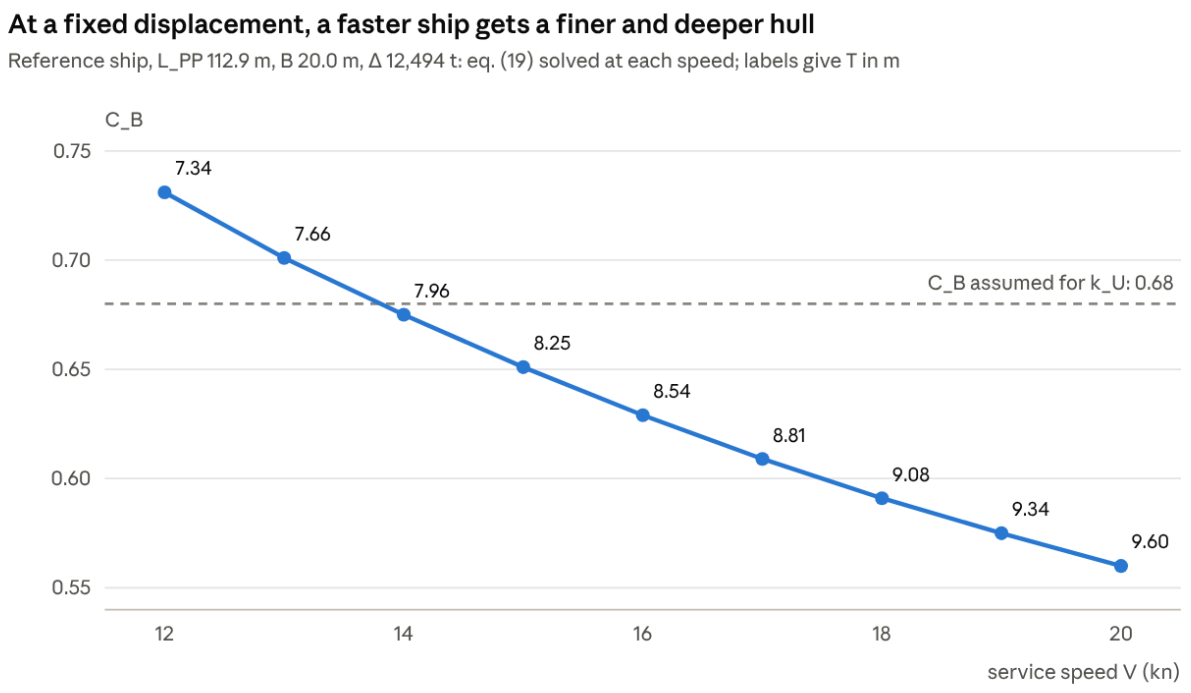


Figure II.11 · C_B and T against speed at fixed displacement, reference ship (sensitivity only)

V	C_B	T	Fn	C_M	C_P
16 kn	0.629	8.54 m	0.247	0.979	0.642
16.5 kn (the ship)	0.619	8.68 m	0.255	0.979	0.632
17 kn	0.609	8.81 m	0.263	0.978	0.623
18 kn	0.591	9.08 m	0.278	0.977	0.605

At the ship's 16.5 kn, Katsoulis gives `C_B` = 0.619 and a draught of 8.68 m, 0.8 m deeper than the real ship. Read the other way round, at the real draught of 7.9 m and 16.5 kn Katsoulis gives `C_B` = 0.609, a displacement of 11,192 t and a lightship of 2,299 t. The two readings cannot both hold: without the published lightship, the block coefficient of the reference ship stays an assumption. Katsoulis returns 0.68 only at about 13.8 kn. At 16–18 kn the same displacement needs a finer hull and 0.6–1.2 m more draught. `C_B`, `V` and `Δ` must come from the same brief.

5.3 The other coefficients

From the Froude number `Fn = V / √(g L_PP)`, with `V` in m/s and `g` = 9.81 m/s², eq. (20), from Eq. 2.9.13:

$$C_M = 1 - 0.062\,Fn^{0.792}, \qquad C_P = C_B / C_M \qquad (20)$$

Check `Fn` against the last column of the ratio table of Step 3 · Main dimensions from the container grid, 3.4 Check the ratios (0.23–0.26 for the smaller Panama ships).

Alvariño et al. take `C_WP` and the longitudinal centre of buoyancy from Part 3 of their work. In this course use the estimates of Lecture 2.3a and the sectional-area curve of homework H5.

5.4 Close the iteration with step 2

The hold utilization `k_U = 0.5 C_B + 0.4` used a starting `C_B`. Recompute `n_B` with eq. (8) and the new `C_B`; if it changes, rebuild `L_PP` and repeat this step (§2.9.3.2). Then check the freeboard and the ratios again, as in 3.4 Check the ratios, with the new `T`; if the freeboard is too small, go back to 3.3 Depth for

more depth. For example, at 17 kn `C_B` = 0.609 gives `k_U` = 0.70 instead of 0.74: the bottom of the range of I.3, where none of the three candidates changes its number of bays (III.1).

Check before moving on.

- The loop converged; `T` is within the draught limit of the brief, and `D − T` leaves the minimum freeboard.
- `V` is the service speed of the brief, stated with the result.
- `k_U` was recomputed with the new `C_B`, and `n_B` checked again.
- `Fn` is inside the range of the ratio table.

Step 6 · Weights and power

Step card	
Purpose	build the lightship from its parts, add the deadweight item by item, and close the balance with the hull
Inputs	`L_PP`, `B`, `D`, `C_B`, `V` (Step 5 · Displacement, C_B and draught); deadweight, range, crew and mass per TEU from the brief
Outputs	the power `P`; `LWT` (steel, deckhouse, outfit, machinery, margin) with its height; the deadweight items; the final `Δ`, `C_B` and `T`
Equations	(21) to (30)
Code	`admiralty_power`, `consumables`, `lightship`, `close_weights`, `close_design` (Appendix D)

Step 5 · Displacement, C_B and draught needs a lightship, and a similar ship rarely gives one. This step builds it from its parts and adds the deadweight item by item, in the order of Alvariño et al. (their Fig. 2.9.8).

Their figure names each estimate only as a function of the main data. The formulas used here are those of Papanikolaou §2.15, for container ships wherever the book gives them.

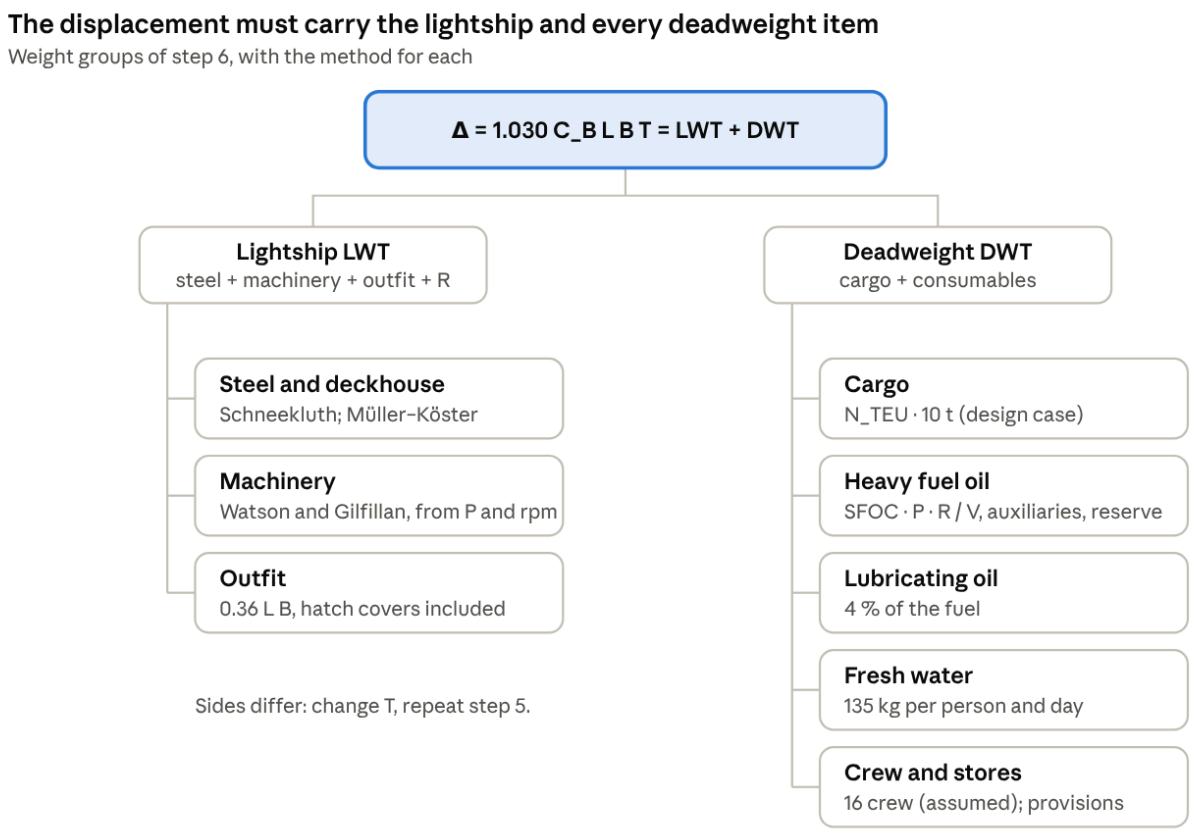


Figure II.12 · Weight groups: lightship and deadweight, with the estimate for each

6.1 Power

At this stage the hull has no lines, so a resistance method cannot be run yet. Take a first estimate of the installed power with the Admiralty formula (Papanikolaou Eq. 2.104), eq. (21):

$$P = \frac{\Delta^{2/3}\, V^3}{C_N} \qquad (21)$$

`C_N` comes from a similar ship. For the reference ship, 6,000 kW at 16.5 kn and the 12,494 t of 5.2 Solve C_B and T give `C_N` = 403. That displacement rests on the assumed `C_B` = 0.68 (I.5), so `C_N` does too.

Because 6,000 kW is the installed rating, `P` includes the margins: it is the power the engine must have, not the power in service. The formula assumes similar hull forms, lengths and speeds. It is good enough to compare candidates and to size the machinery and the fuel; it is not a power prediction.

Before a design is accepted, power it with a resistance and propulsion method. In this course that is Holtrop and Mennen (Lecture 2.4, Tool 2.4 and Demo 5), which needs the hull form of 5.3 The other coefficients. The propeller selection of homework H2b then refines the open-water efficiency. Replace `P` by that result and close the balance of 6.6 Close the balance again.

6.2 Deadweight items

Work through them in the order of their Fig. 2.9.8. The rates are those of Papanikolaou §2.15.7; the range and the crew come from the brief. The fuel for the round voyage, with `R` its length in nautical miles, eq. (22):

$$W_F = P \cdot SFOC \cdot \frac{R}{V}\,(1 + a_{aux})\, C \cdot 10^{-6} \qquad (22)$$

Item	How to estimate (Papanikolaou §2.15.7)
Fuel	eq. (22): `SFOC` = 190 g/kWh for a medium-speed diesel; the auxiliary engines burn `a_aux` = 5–7 % of the main engine's fuel; the reserve `C` = 1.2–1.4 covers detours, waiting and tank residues
Lubricating oil	3–5 % of the fuel
Fresh water	per person and day, 10–20 kg for drinking and 120 kg for washing, over the days at sea
Provisions	7–16 kg per person and day
Crew and effects	75 kg per person, plus 60 kg of luggage
Cargo	`N_TEU` · 10 t per TEU (design); 14 t per TEU in the homogeneous condition (7.4 Loading conditions)

Watch the units in eq. (22): `SFOC` in g/kWh times `P` in kW times hours gives grams, hence the factor 10^{-6} for tonnes. This manual uses `a_aux` = 0.06, `C` = 1.3, lubricating oil 4 %, 135 kg of water and 11.5 kg of provisions per person and day.

Example: the brief of the reference ship. Its range and crew are not published. This manual assumes a round voyage of 3,000 nm and a crew of 16; they are assumptions of the example, not data.

For the first candidate of III.1, `P` = 6,148 kW and the voyage takes 3,000 / 16.5 = 182 h, or 7.6 days. The items are: fuel 293 t, lubricating oil 12 t, water 16 t, provisions 1.4 t and crew 2.2 t, 324 t in all.

Which deadweight. The brief of the example fixes the deadweight, 8,893 t, as Alvariño et al. assume: the cargo is what is left after the consumables (their box 21). Here that is 8,569 t, more than the 5,960 t of the design cargo; the rest is room for ballast (7.4 Loading conditions).

In the project the brief fixes the cargo instead, `N_TEU` · 10 t. Its first `DWT` = cargo / 0.75 (Fig. A.49), used in 5.1 First displacement, is then replaced by the sum of the items.

6.3 Steel

The lightship is split in three parts, each with its own section: steel (6.3 Steel), machinery (6.4 Machinery) and outfit (6.5 Outfit).

Estimate the hull steel by Schneekluth's formula for container ships (Papanikolaou §2.15.4), eq. (23). It was fitted for `L` = 100–250 m, `L/D` = 10–15.5, `B/D` = 1.47–2.38 and `C_B` = 0.52–0.716, which covers the candidates. It includes the container cell guides, but not the deckhouse and the hatch covers.

$$W'_{ST} = \nabla_U \, C'_{ST} \left[1 + 2 \cdot 10^{-6}(L-120)^2\right] \left[1 + 0.057\left(\frac{L}{D} - 12\right)\right] \sqrt{\frac{30}{D+14}} \left[1 + 0.1\left(\frac{B}{D} - 2.1\right)^2\right] \left[1 + 0.2\left(\frac{T}{D} - 0.85\right)\right] \left[0.92 + (1 - C_{BD})^2\right]$$

$$\nabla_U = L\,B\,D\,C_{BD} + n_{HO}\,l_{hold}\,(B - 2\,w_{DH})\,h_{CO}, \qquad C_{BD} = C_B + C_1 \frac{D-T}{T}\,(1 - C_B) \tag{23}$$

`C'_ST` = 0.090–0.100, 0.093 on average; this manual uses 0.093. `∇_U` is the volume under the deck plus the hatch coamings (Papanikolaou Eq. 2.115). With no sheer and no camber it is the hull up to `D`, plus coamings one hold long and as wide as the space between the wing tanks.

`C_BD` is the block coefficient at the depth `D`, with `C_1` = 0.25 for sections of small flare. Schneekluth's corrections for mild steel only, for a trapezoidal midship section and for a raised double bottom are not applied.

The same method gives the height of this steel above the base line, eq. (24). With no sheer, the corrected depth `D_S` of the book equals `D`:

$$KG'_{ST} = \left[44 + 0.155\,(0.85 - C_{BD})\left(\frac{L}{D}\right)^2\right]\frac{D}{100} \qquad (24)$$

Its small corrections, for transverse framing, a bulbous bow, `L/B` away from 6.5 and `L` away from 120 m, are not applied either.

The deckhouse. Add it by Müller–Köster (Papanikolaou §2.15.4), eq. (25). Each layer of the deckhouse weighs a volumetric coefficient `C_DH` times its volume:

$$W_{DH} = \sum_j C_{DH,j}\, l_{DH}\, b_{DH}\, h\, k_1\, k_3, \qquad k_1 = 1 + 0.02\,(h - 2.6) \qquad (25)$$

The reference ship's deckhouse stands aft over the engine room and is 12 m long (I.5). Its breadth and number of layers are not published. This manual assumes the full beam and five layers, I to IV and the wheelhouse, filling the height from the deck to the bridge deck of eq. (15).

With no outside walkways, `A_O/A_U` = 1.0, so `C_DH` = 57, 55, 52, 53 and 40 kp/m³ (Table 2.26). `k_3` corrects for short ships: 0.95 at 100 m, interpolated to 1.0 at 120 m; `k_2` = 1 for standard internal walls.

Cross-check: Watson and Gilfillan. Papanikolaou also gives Watson and Gilfillan's formula with `K` = 0.036 for container ships, eqs. (26) and (27). `C_B'` is the block coefficient corrected to 80 % of the depth. In eq. (27), `l_1` and `h_1` are the length and height of each superstructure, and `l_2` and `h_2` those of each deckhouse:

$$W_S = K\,E^{1.36}\,[1 + 0.5\,(C'_B - 0.70)]\,, \qquad C'_B = C_B + (1 - C_B)\,\frac{0.8\,D - T}{3\,T} \qquad (26)$$

$$E = L\,(B + T) + 0.85\,L\,(D - T) + 0.85\sum l_1 h_1 + 0.75\sum l_2 h_2 \qquad (27)$$

Check `E` before you trust `K`: it was fitted for `E` = 6,000–13,000. The candidates of III.1 have `E` = 4,339–4,772, so eq. (26) serves only as a cross-check. For the first candidate it gives 3,101 t of steel, 24 % above the 2,509 t of eqs. (23) and (25).

6.4 Machinery

Estimate the machinery by Watson and Gilfillan (Papanikolaou §2.15.6), eq. (28). The main engine follows from its rating `P` (kW) and speed `n` (rpm); the rest of the machinery from the rating alone, with 0.69 for container ships:

$$W_M = 12\,\frac{P}{n} + 0.69\,P^{0.70} \qquad (28)$$

The reference ship has a medium-speed diesel at 500 rpm, and the candidates keep that type of engine. Once an engine is chosen, take its dry mass from the maker's catalogue instead.

6.5 Outfit

Estimate the outfit from the deck area, eq. (29), with `K_OT` = 0.34–0.38 t/m² for container ships (Papanikolaou §2.15.5); this manual uses 0.36. The outfit includes the hatch covers and the lashing gear on deck.

$$W_{OT} = K_{OT}\,L\,B, \qquad LWT = W'_{ST} + W_{DH} + W_{OT} + W_M + R \qquad (29)$$

`R` is a margin of 1–3 % of the other groups (§2.15.1); this manual uses 2 %. The guide weights of the footnote of Table 2.9.2 are not used (Appendix B).

6.6 Close the balance

The displacement must carry what you estimated, eq. (30):

$$\Delta = 1.030\,C_B\,L\,B\,T = LWT + DWT \qquad (30)$$

This is the weight balance of Lecture 3.1: lightship plus deadweight against the hull. In Lecture 3.1, `C_B` is held and the shortfall divided by the TPC gives the change of draught.

Here `C_B` follows the draught through Katsoulis, eq. (19), and the lightship follows `T`, `C_B` and the power. So the balance is repeated:

1. Start from a draught and a `C_B`.
2. Compute the power, eq. (21), and the lightship, eqs. (23) to (29).
3. Solve eq. (19) for the displacement `LWT + DWT`: this gives a new `C_B` and `T`.
4. Repeat until `T` changes by less than 1 cm. The shortfall is then well inside the margin `R`.

Example: the first candidate of III.1 (`s_H` = 0.45, 7 rows, 4 tiers, 14 bays: `L_PP` = 128.4 m, `B` = 22.05 m, `D` = 11.33 m), with `DWT` = 8,893 t at 16.5 kn:

Pass	`T` in	`C_B` in	`P`	`LWT`	`LWT + DWT`	Hull, 1.030 `C_B L B T`	Shortfall	`T`, `C_B` out
1	7.60 m (`B` / 2.9)	0.680	6,799 kW	4,204 t	13,097 t	15,072 t	−1,976 t	7.30 m, 0.616
2	7.30 m	0.616	6,191 kW	4,070 t	12,963 t	13,097 t	−134 t	7.23 m, 0.615
3	7.23 m	0.615	6,149 kW	4,065 t	12,958 t	12,963 t	−4 t	7.23 m, 0.615

A negative shortfall means the hull floats more than the ship weighs, so the draught comes down. After three passes the two sides differ by 4 t, against a margin `R` of 80 t.

The lightship of pass 3 is 2,306 t of hull steel, 203 t of deckhouse, 1,019 t of outfit, 457 t of machinery and an 80 t margin. Then `k_U` = 0.5 × 0.615 + 0.4 = 0.71 still gives 14 bays, so 5.4 Close the iteration with step 2 does not start the loop again.

Why only the draught moves. In this method the hull cannot be scaled to the weights. Length, beam and depth are built from whole bays, rows and tiers, eqs. (9) to (11), so a hull 3 % wider would need 7.2 rows. Once the grid is fixed, the draught and the block coefficient are the only quantities left to float the weights.

Papanikolaou (2026) counts container ships among the volume carriers (§1.3.7.2), and calls ships whose dimensions are fixed by box-shaped cargo linear dimension ships (§1.3.7.3). For volume carriers his sequence ends with `Δ = W_L + DWT`, and only then the draught, from `Δ`, `L`, `B` and `C_B` (Table 1.9). The draught is a result, not a choice.

A deadweight carrier, such as the general cargo ship of the course lectures, goes the other way (Table 1.8). The displacement comes first, from the deadweight. Length, beam and draught then follow from `Δ` and the ratios, so the whole hull grows or shrinks with the weights.

	Deadweight carrier (Table 1.8)	Container ship, this manual (Table 1.9; Alvariño et al. §2.9.3)
Starts from	the deadweight, then `Δ`	the TEU, then the grid
`L`, `B`, `D`	from `Δ` and the ratios; any value	from bays, rows and tiers; in steps of one container
`Δ`	a first estimate (Table 2.1)	a result: `LWT + DWT`
Draught	from `B/T` and `L/T`, early	last, from `Δ`, `L`, `B` and `C_B` (eq. 19)

	Deadweight carrier (Table 1.8)	Container ship, this manual (Table 1.9; Alvariño et al. §2.9.3)
When the weights change	scale the hull	change `T` and `C_B`; if `T` fails, take another grid
Ratios	used to size the hull	used to check it

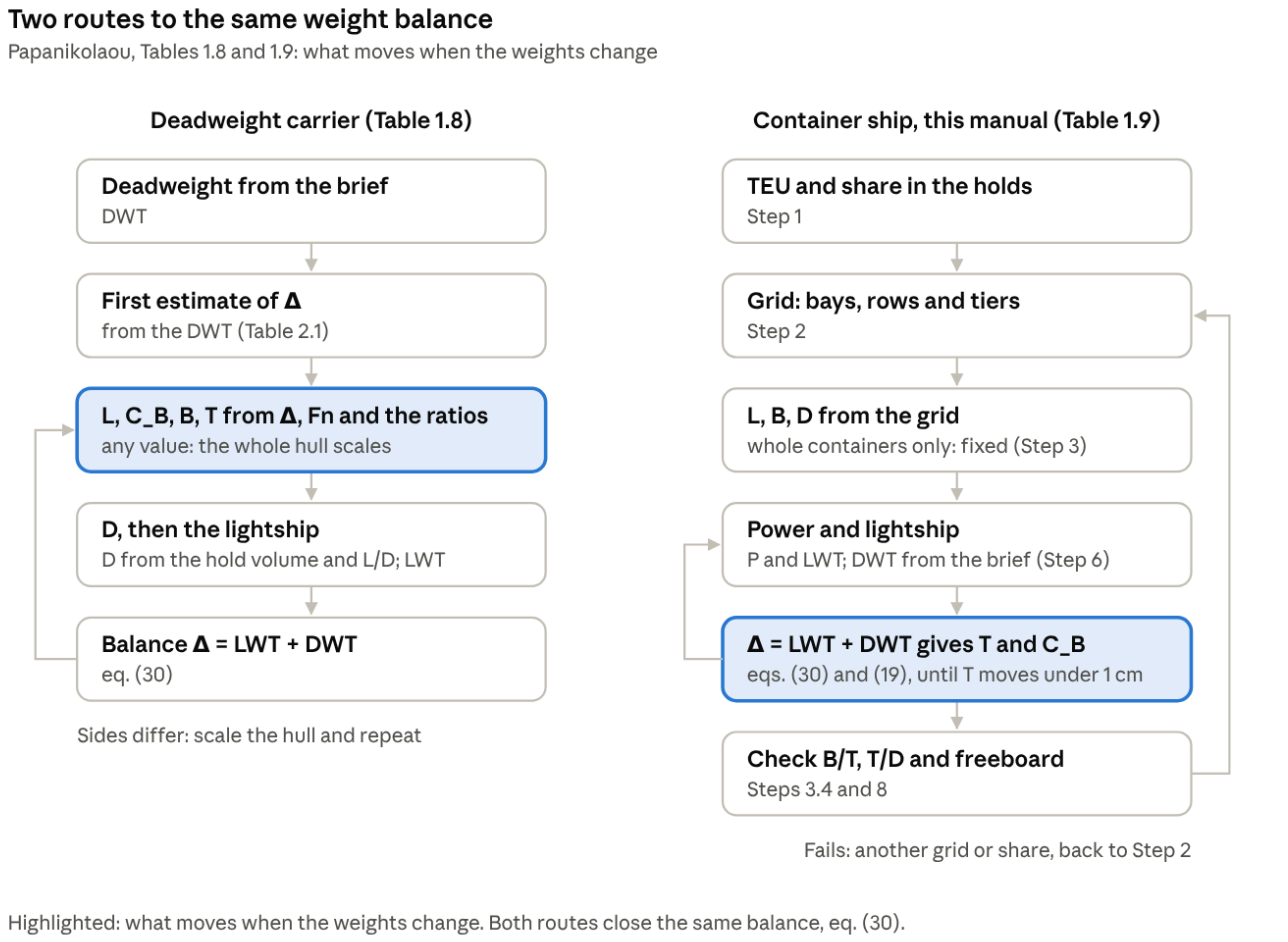


Figure II.13 · Two routes to the weight balance: deadweight carrier (Papanikolaou Table 1.8) and container ship (Table 1.9)

Both routes close the same balance, eq. (30), as Figure II.13 shows. Alvariño et al. also allow keeping `C_B` and adjusting only the draught, with their Eq. 2.9.12 (§2.9.3.2), which is the route of Lecture 3.1.

If the draught that balances the weights does not suit the hull, the hull is not stretched. `D` is fixed by the tiers, so a deeper draught eats the freeboard; a shallower one moves `B/T` and `T/D` out of range. The answer is another grid or share, which the sweep already contains (8.2 What to change when a check fails).

In III.1 this is what rejects the two wider or longer candidates: with 8,893 t of deadweight they float at only 6.7–7.0 m, too shallow for their beam and depth.

Record the heights too. Step 7 · Stability needs the height of every weight above the base line. Eq. (24) gives the hull steel, and the deckhouse stands at half its height above the deck.

For the outfit and the machinery, take the middle of the ranges for container ships in Table 2.19 of Papanikolaou: 86–105 % and 29–53 % of `D_S`. `D_S` is the depth corrected for the deckhouse volume, `D + ∇_DH / (L B)` (Eq. 2.113). The table starts at 10,000 t of deadweight; smaller ships have their centres higher.

Put the margin `R` 20 % above the mean height of the other groups, as in Lecture 3.1. For the first candidate this gives `D_S` = 12.71 m and a lightship `KG` of 7.81 m.

Check before moving on.

- Every item of Figure II.12 has a mass, a source and a height.
- The power is marked as an Admiralty estimate until a resistance method replaces it.
- The steel formula is inside its range of application; eq. (26) is used only as a cross-check.
- The shortfall of eq. (30) is inside the margin `R`.
- The final `T` leaves the minimum freeboard, and the ratios of 3.4 Check the ratios still pass.

Step 7 · Stability

Step card	
Purpose	check that each candidate is stable with its deck load and the ballast it needs, and find how many TEU of 14 t stability allows
Inputs	the weights and heights of Step 6 · Weights and power; `T`, `C_B`, `B`, `L_PP`; the grid and `n_TD`
Outputs	`KG`, `GM`, ballast and roll period for each loading condition; the homogeneous capacity
Equations	(31) to (36)
Code	`tier_heights`, `centre_of_gravity`, `stability`, `ballast_needed`, `homogeneous_capacity` (Appendix D)

Alvariño et al. name stability as the main design problem of container ships: they have no freeboard problem, but many tiers stand on the hatch covers, and they often sail with the double-bottom tanks ballasted (§2.9.3.0). Their Fig. 2.9.10 computes the height of the centre of gravity above the base line, `KG`, the metacentric height `GM` and the roll period. This is the step where the deck stacks of Step 4 · Capacity on deck and visibility meet their second limit (Figure I.3).

7.1 Height of the cargo

Take each container's centre of gravity at half its height; statistically it lies at 45 % of the height (§2.9.2), so mid-height is on the safe side. For tier `i`, counted from the bottom, with the tier pitch `p_T = h_c + 0.013`, the vertical spacing of the tiers, eq. (31):

$$z_{hold,i} = h_{DB} + (i-1)\,p_T + 0.5\,h_c, \qquad z_{deck,i} = D + h_{CO} + t_{cover} + (i-1)\,p_T + 0.5\,h_c \tag{31}$$

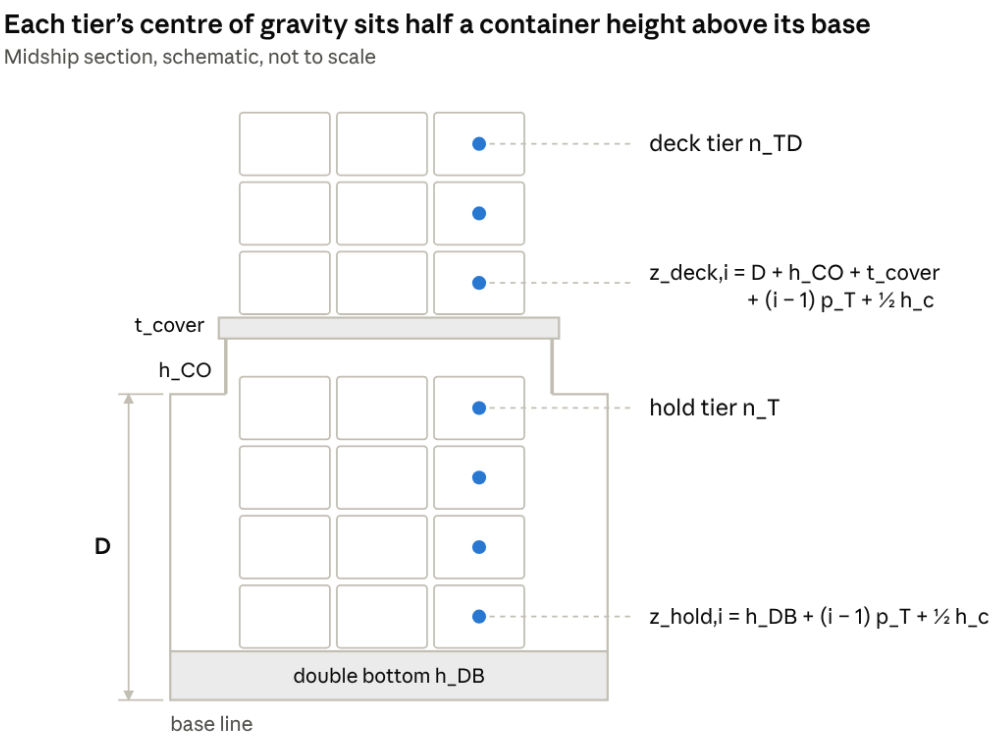


Figure II.14 · Height of each tier's centre of gravity, eq. (31) (schematic)

Example: the reference ship. High cubes (`h_c` = 2.896 m, `p_T` = 2.909 m), `h_DB` = 0.8 m, `D` = 11.3 m, `h_CO` = 1.6 m, `t_cover` = 0.6 m, so the deck tiers stand on 11.3 + 1.6 + 0.6 = 13.5 m:

Tier	1	2	3	4
Hold, m above base	2.25	5.16	8.07	10.98
Deck, m above base	14.95	17.86	20.77	23.68

The fourth deck tier sits more than twice as high as `D`. Every deck tier adds mass far above the ship's `KG`, which is why eq. (36) below grows quickly with each tier.

7.2 `KG` of the ship

Add the lightship of Step 6 · Weights and power at the height found in 6.6 Close the balance, the deadweight items, the ballast and the cargo of 7.1 Height of the cargo. With masses `m_j` at heights `z_j` above the base line, eq. (32):

$$KG = \frac{\sum m_j \, z_j}{\sum m_j} \qquad (32)$$

Put the tanks and the ballast in the double bottom at half its height, and the provisions and crew in the deckhouse. Both are assumptions of this manual: the tank plan comes later.

7.3 `KB`, `BM` and `GM`

Take the height of the centre of buoyancy, `KB`, by Morrish's formula, and the metacentric radius `BM` as the transverse inertia of the waterplane, `I_T`, over the displaced volume `∇` (Papanikolaou §2.18). `δGM_FS` is the correction for the free surface of partly filled tanks, eq. (33):

$$KB = T\left(\frac{5}{6} - \frac{C_B}{3\,C_{WP}}\right), \qquad BM = \frac{I_T}{\nabla}, \qquad GM = KB + BM - KG - \delta GM_{FS} \qquad (33)$$

Without lines, estimate `C_WP` and `I_T` from `C_B`, as in Lecture 2.3a (Papanikolaou Eqs. 2.83 and 2.95). For each loading condition, take the draught from its displacement with `C_WP` held, as Demo 5 does, and `C_B` at that draught.

The course texts give no preliminary estimate of the free surface. This manual assumes one slack tank in every condition: a double-bottom fuel tank one hold long and half the inner bottom wide, as if the tanks were split at the centre girder. With `ρ_F` = 0.95 t/m³ (heavy fuel oil, 0.92–1.02 t/m³, §2.15.7), eq. (34):

$$C_{WP} = \frac{1 + 2\,C_B}{3}, \qquad I_T = C_{WP}\,(0.1272 + 0.8724\,C_{WP})\,\frac{L\,B^3}{12}, \qquad \delta GM_{FS} = \frac{\rho_F\,l_t\,b_t^3}{12\,\Delta} \tag{34}$$

`BM` grows with `B²`, while `KG` grows with every tier: of the four grids of Step 2 · Lay out the containers, the extra row works for stability and the extra tier against it. In III.1 this is what separates the 7-row candidates, which need ballast, from the 8-row one, which does not.

7.4 Loading conditions

Run these three. Stow the cargo in the holds first, the same number of TEU in each tier, then on deck from the lowest tier up:

Condition	Cargo	Consumables	What it gives
Design, departure	all slots at the design mass per TEU of the brief (10 t)	100 %	`GM`, with the ballast it needs
Design, arrival	as above	10 %	`GM`, with the ballast it needs
Homogeneous	14 t per TEU, as many as stability and the deadweight allow (7.7 What to change)	100 %	the number of TEU

The arrival condition is usually the critical one: the fuel and water burnt were low in the ship, so `KG` rises.

Ballast. Container ships often sail with the double bottom ballasted to get enough `GM` (Alvariño et al. §2.9.3.0; Papanikolaou §2.15.7). If `GM` is below the minimum of 7.5 Criteria, add water ballast in the double bottom, in steps of 10 t, until it is reached.

The ballast is limited twice: by the deadweight left after the cargo and the consumables, and by the volume of the double bottom under the cargo zone, less its fuel and water. This manual takes that volume as a box, `6.55 n_B (B − 2 w_DH) h_DB`. If that ballast is not enough, the candidate fails the `GM` check.

7.5 Criteria

Papanikolaou §2.18.8 lists the criteria of the IMO Intact Stability Code (IS Code): five general criteria (a to e), the weather criterion, and, for container ships longer than 100 m, criteria on the GZ curve scaled by a form factor of the hull. Only criterion (e) can be checked with the results of this step: the initial `GM`, corrected for free surfaces, not less than 0.15 m. All the others need the righting-lever (GZ) curve, which needs the hull lines; they are checked later, once the lines exist. So check `GM` ≥ 0.15 m in each loading condition of 7.4 Loading conditions.

A target, not a criterion: `GM` ≥ 0.06 `B`. Papanikolaou (§2.12) says that `GM` should, in general, not be smaller than about 0.06 `B`, and the course's synthesis model (Demo 5) uses that value. Report it for each candidate, but do not reject a design on it here.

Container ships are often limited by stability rather than by space. The reference ship is published with 620 TEU, but only 410 TEU of 14 t (I.5). That is well below what its deadweight would carry, which points to stability or stack weights as the limit.

7.6 Roll period

Compute the natural roll period with the IS Code estimate, eq. (35), with `L_WL` the waterline length and `T` the draught. Without lines, this manual takes `L_WL` = `L_PP`:

$$T_\phi = \frac{2\,C\,B}{\sqrt{GM}}, \qquad C = 0.373 + 0.023\,\frac{B}{T} - 0.043\,\frac{L_{WL}}{100} \qquad (35)$$

The IS Code sets no limit on the roll period. Papanikolaou (Table 2.13) gives 12–18 s as typical for cargo ships. A large `GM` gives a short period and high accelerations on the deck stacks and their lashings, so this manual checks only the lower end: `T_φ` ≥ 12 s.

A long period shows a small `GM`, which 7.5 Criteria already checks. If the roll check fails, `GM` is too large: go back to 3.2 Beam or 3.3 Depth. The target is enough `GM`, not the most.

7.7 What to change

When the `GM` or the roll check fails, go back to the beam or the depth (3.2 Beam or 3.3 Depth). On deck, the number of TEU that can actually be carried depends on stability and is usually below the geometric count of Step 4 · Capacity on deck and visibility (§3.8.6.2). By eq. (36), one more deck tier of mass `m_D` at mean height `z̄_D` raises `KG` by:

$$\delta KG = \frac{m_D\,(\bar{z}_D - KG)}{\Delta + m_D} \qquad (36)$$

Homogeneous capacity. The homogeneous condition of 7.4 Loading conditions measures this limit. Load TEU of 14 t, holds first, and add the ballast the ship has room for; the most TEU that keep `GM` ≥ 0.15 m is the homogeneous capacity. Compare it with the brief: the reference ship is published with 410 TEU of 14 t (I.5).

Check before moving on.

- The heights of eq. (31) start from `h_DB` in the hold and from `D + h_CO + t_cover` on deck.
- Every mass of Step 6 · Weights and power has a height, and `KG` was computed for all three conditions.
- `GM`, after the free-surface correction, is at least 0.15 m in departure and arrival, with the ballast stated; `GM` against 0.06 `B` is reported.
- The roll period is at least 12 s, and the homogeneous capacity is stated.

Step 8 · Check and iterate

Step card	
Purpose	decide which candidates are acceptable, and send the others back to the right step
Inputs	every candidate with the results of Step 2 · Lay out the containers to Step 7 · Stability
Outputs	the domain of solutions: the candidates that pass all five checks, with the weight balance closed
Equations	none new; the checks reuse (8), (12), (19), (30), (33), (35)
Code	the sweep loop, `size_grid` (Appendix D)

8.1 The five checks

A candidate is accepted when the five checks pass together, after the weight balance has been closed. The table says what each check tests, in which section it is made, and where to go back when it fails.

Check	What it tests	Made in	If it fails, go back to
Freeboard	the depth leaves at least the ICLL basic freeboard of a Type B ship (Papanikolaou Table 2.42, without its corrections) above the draught, `D − T`	3.4 Check the ratios and 5.4 Close the iteration with step 2	3.3 Depth, the depth
Ratios	`L/B`, `L/D`, `B/D`, `B/T`, `T/D` inside the app. 2.9 ranges, and Fn inside its range, at the draught that balances the weights	3.4 Check the ratios and 5.4 Close the iteration with step 2	3.1 Length to 3.3 Depth (length, beam, depth) or 5.2 Solve C_B and T (draught)
Capacity	TEU in the holds, on deck and in total	2.4 Bays and 4.1 Containers on deck	Step 2 · Lay out the containers, another grid or share
`GM`	the metacentric height, corrected for free surfaces, is at least 0.15 m in departure and arrival, with the ballast the double bottom can take	7.5 Criteria	3.2 Beam or 3.3 Depth, the beam or the depth
Roll period	the roll period is at least 12 s (Papanikolaou Table 2.13)	7.6 Roll period	3.2 Beam or 3.3 Depth, the beam or the depth

The weight balance `Δ = LWT + DWT` (6.6 Close the balance) is not a check: it is closed by adjusting the draught in 5.2 Solve C_B and T until the two sides agree. The candidate is dropped only if the new draught then fails the freeboard or ratio check.

8.2 What to change when a check fails

In this method the dimensions are not free numbers, they come from whole containers. So a failure is answered by a different count, which means a neighbouring grid of Step 2 · Lay out the containers or another share. The table says which one to look at; inside the sweep that candidate already exists, outside it add it as a new candidate:

Failure	First thing to change	Effect
Too long and shallow (`L/D`, `B/D`)	one more tier, `n_T + 1`	`D` + 2.9 m, fewer bays
Too slender (`L/B`)	one more row, `n_R + 1`	`B` + 2.5 m, fewer bays
Too short and beamy (`L/B` low)	one row fewer, or a lower share `s_H`	more bays
Deck needs more than 5 tiers, or the sight line cuts too much	a higher share `s_H`, or a higher bridge	fewer TEU on deck
`T` above the draught limit of the brief, or `D − T` below the minimum freeboard	review the weights; then `n_T + 1` (freeboard) or `n_R + 1` (draught limit)	more depth; or more beam and less draught
`GM` too small	one more row, or fewer deck tiers	higher `BM`, or lower `KG`

The technical evaluation that follows (the last chart of Figure II.1) has its own checks: freeboard, TEU count, manoeuvring, lightship, trim, stability, strength, arrangement and price. They come after this manual.

8.3 From candidates to a domain

Alvariño et al. allow other combinations of bays, rows and tiers whose product satisfies Eq. 2.9.4, each sized with the same equations and the lightship scaled by Eq. 2.9.17, and then compared with a merit function (§2.9.3.2); they also note that in practice only one is usually studied.

This manual studies them all: five shares `s_H` (Step 1 · Split the TEU and set targets) times four grids (Step 2 · Lay out the containers), 20 candidates. The candidates that pass every check of 8.1 The five checks form the domain of solutions of Lecture 2.3a; the economic evaluation of the project picks the best of them. III.1 runs this sweep on the reference ship.

Check before moving on.

- All five checks pass for each candidate you keep, with the weight balance closed.
- For each rejected candidate you can name the check that failed.
- The kept candidates, with their dimensions, `C_B`, `T`, `Δ` and `GM`, are listed in one table: that table is the hand-over to the economic evaluation.

Part III · Application

Part III runs the method on the reference ship of I.5 and then tests its estimates against two built ships. Every number comes from the lecturer's program, built from the functions of Appendix D. Write your own program and check that it reproduces them before you use it for your own brief.

III.1 Worked example · the 596 TEU feeder

Step 1 · Split the TEU and set targets to Step 8 · Check and iterate are run as if the reference ship were being designed from its 596 TEU, and the result is compared with the real ship. Its lightship is not published, so Step 6 · Weights and power builds one from its parts; its range and crew are not published either, and are assumed (6.2 Deadweight items).

Inputs.

Input	Value	Where from
`N_TEU`, `s_H`	596 TEU, 0.40	brief; middle of 0.30–0.50 (§2.9.3.1)
`w_DH`, `h_DB`, `h_CO`, `c_HC`, `c_DK`	2.1, 0.8, 1.6, 0.49, 0 m	the reference ship (I.5)
`g_BC`, `g_BF`	0.15, 0.08 m	Table 2.9.2, lower values
`L_AP + L_ER`, `L_FP`	21.6, 15.1 m	the reference ship
`C_B` for `k_U`	0.68, so `k_U` = 0.74	assumed for the reference ship (I.3)
`n_l`, container	2 bays over the engine room; high cube	the reference ship

Step 1. `N_H` = 238 and `N_D` = 358 TEU by eq. (1). Eqs. (2) and (3) give `L_0` = 117.9 m, `B_0` = 20.44 m, `D_0` = 9.97 m and `T_0` = 7.58 m (freeboard draught). Fig. A.47 gives 116.0 m for 596 TEU; `L_0` is 1.6 % above it. The reference ship's draught, 7.9 m (Step 5 · Displacement, C_B and draught), is above `T_0`: `T_0` is a target, not a limit.

Steps 2–3. Rows by eq. (7): ⌊(20.44 − 4.2) / 2.54⌋ = 6. Tiers by eq. (7): ⌊(9.97 + 1.6 − 0.8 − 0.49) / 2.909⌋ = 3. `c_HC` = 0.49 m is 4 % of the 12.19 m hatch cover of a 40 ft bay (Eq. 2.9.3); with it, eq. (11) gives 11.33

m for four tiers, against the real ship's 11.3 m. Bays by eq. (8), dimensions by eqs. (9)–(11), then the ratio check decides. Three of the four grids of Step 2 · Lay out the containers are named candidates here:

Candidate	n_R	n_T	n_B	L_PP	B	D	Ratios outside the range (< 1,000 TEU row)
A, grid 1	6	3	18	154.6 m	19.53 m	8.42 m	L/B 7.92, B/D 2.32, L/D 18.4, T/D 0.90
B, grid 2, one more tier	6	4	14	128.4 m	19.53 m	11.33 m	L/B 6.58 (limit 6.5)
C, grid 4, one more row	7	4	12	115.3 m	22.05 m	11.33 m	L/D 10.18 (limit 10.5)

Candidate A fails because D_0 comes from ships with 8 ft 6 in containers: it leaves room for only three high-cube tiers, so the ship grows long and shallow. The ratio check of 3.4 Check the ratios sends it back to the depth; one more tier gives B. B is just above the L/B range; the check sends it back to the beam, and one more row gives C. C is just below the L/D range: with 12 bays it is short for its depth. No grid of this share passes, so the designs come from the other shares (the sweep, below). (Grid 3, one more row only, fails like A: L/D 16.8.)

Step 4, on the first candidate that passes step 3. At s_H = 0.45 (N_H = 268, N_D = 328), grid 4 has 7 rows × 4 tiers × 14 bays: L_PP = 128.4 m, B = 22.05 m, D = 11.33 m. Deck by eq. (12): (14 + 2) · (7 + 2) · n_TD ≥ 328 needs n_TD = 3, which carries 432 TEU. Sight line by eqs. (13)–(15): with the deckhouse over the engine room (eye 10.8 m from the aft perpendicular, 27.9 m above base), every bay can carry at least 3 tiers. No TEU are lost, and the deck has 104 TEU of margin.

Against the real ship. The real ship has the rows and tiers of candidate B, but 11 bays instead of 14. The difference is the hold utilization: the formula gives k_U = 0.74; the real ship holds 228 of its 264 grid cells, k_U = 0.86 (I.3). With 0.86, candidate B needs 12 bays, one pair more than the real ship. Calibrate k_U on your reference ship. The real ship itself is just outside the L/D range: 112.9 / 11.3 = 9.99, below 10.5.

The sweep. Candidates A–C use one share, s_H = 0.40. Running the five shares with four grids each (the sweep loop of Appendix D) gives 20 candidates. Seventeen fail the ratio check: every grid with three tiers, which comes out too long and shallow; the four-tier grids with too many bays for their beam, too slender (L/B); and those with 12 bays, 115.3 m, too short for their depth (L/D 10.18). None fails the deck check of Step 4 · Capacity on deck and visibility.

The sweep: 20 candidates, 3 pass the ratio and deck checks

Reference ship brief, 596 TEU, k_U 0.74: rows × tiers × bays and L_PP of each candidate

	s_H = 0.30	s_H = 0.35	s_H = 0.40	s_H = 0.45	s_H = 0.50
Grid 1 rounded down	5 × 3 × 18 154.6 m	6 × 3 × 16 141.5 m	6 × 3 × 18 154.6 m	6 × 3 × 22 180.8 m	7 × 3 × 20 167.7 m
Grid 2 +1 tier	5 × 4 × 14 128.4 m	6 × 4 × 12 115.3 m	6 × 4 × 14 128.4 m	6 × 4 × 16 141.5 m	**7 × 4 × 16 141.5 m**
Grid 3 +1 row	6 × 3 × 14 128.4 m	7 × 3 × 14 128.4 m	7 × 3 × 16 141.5 m	7 × 3 × 18 154.6 m	8 × 3 × 18 154.6 m
Grid 4 +1 row, +1 tier	6 × 4 × 12 115.3 m	7 × 4 × 12 115.3 m	7 × 4 × 12 115.3 m	**7 × 4 × 14 128.4 m**	**8 × 4 × 14 128.4 m**

Outlined: passes. Three-tier grids are too long for their depth; four-tier grids fail L/B, or L/D at 115.3 m.

Figure III.1 · The sweep on the reference ship brief: five shares, four grids each

The three that pass Step 3 · Main dimensions from the container grid and Step 4 · Capacity on deck and visibility:

s_H	N_H	Rows × tiers × bays	L_PP	B	D	n_TD needed, eq. (12)	Tiers at the foremost bay, eqs. (13)–(15)
0.45	268	7 × 4 × 14	128.4 m	22.05 m	11.33 m	3	3
0.50	298	7 × 4 × 16	141.5 m	22.05 m	11.33 m	2	3
0.50	298	8 × 4 × 14	128.4 m	24.56 m	11.33 m	2	3

The real ship carries 38 % in the holds, between the first two shares. Its grid of 6 rows and 4 tiers fails at both: with 12 bays (`s_H` = 0.35) on `L/D`, with 14 bays (`s_H` = 0.40) on `L/B`.

All three candidates keep their full deck stacks under the sight line at the foremost bay. They go on to Step 5 · Displacement, C_B and draught to Step 8 · Check and iterate, below.

Sensitivity to `k_U`. The three candidates, re-run with `k_U` from 0.70 to 0.86 (I.3):

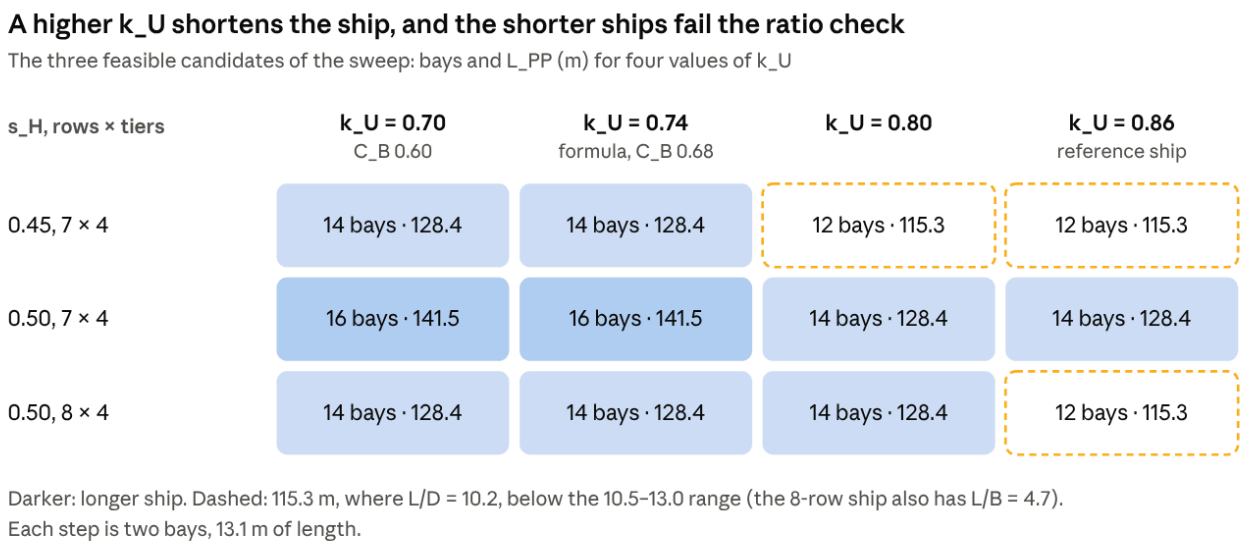


Figure III.2 · Sensitivity of the three candidates of the sweep to k_U, 0.70 to 0.86

s_H	Rows × tiers	k_U 0.70	0.74 (formula)	0.80	0.86 (reference ship)
0.45	7 × 4	14 bays, 128.4 m	14 bays, 128.4 m	12 bays, 115.3 m	12 bays, 115.3 m
0.50	7 × 4	16 bays, 141.5 m	16 bays, 141.5 m	14 bays, 128.4 m	14 bays, 128.4 m
0.50	8 × 4	14 bays, 128.4 m	14 bays, 128.4 m	14 bays, 128.4 m	12 bays, 115.3 m

A higher `k_U` shortens each candidate by two bays, 13.1 m, somewhere in the range; 0.70 changes none of them. A changed length must pass the ratio check again: at 115.3 m the ships have `L/D` = 10.18, below the 10.5–13.0 range, and the 8-row ship also `L/B` = 4.69, below 5.2. The 0.45 candidate fails from `k_U` = 0.80 and the 8-row one at 0.86; only the 0.50, 7 × 4 candidate passes over the whole range. So a higher `k_U` does not simply give a better ship. A candidate whose length jumps inside the range you believe for `k_U` is the one to check most carefully once hull lines exist.

Steps 5 to 8 on the three candidates. Each candidate is closed by the balance of 6.6 Close the balance, with `DWT` = 8,893 t at 16.5 kn, then loaded as in 7.4 Loading conditions and checked as in 8.1 The five checks:

	0.45, 7 × 4 × 14	0.50, 7 × 4 × 16	0.50, 8 × 4 × 14
`L_PP` × `B` × `D`	128.4 × 22.05 × 11.33 m	141.5 × 22.05 × 11.33 m	128.4 × 24.56 × 11.33 m
`T`; `C_B`	7.23 m; 0.615	6.68 m; 0.631	6.95 m; 0.590
`Δ`; `P`, eq. (21)	12,958 t; 6,148 kW	13,553 t; 6,335 kW	13,339 t; 6,268 kW
`LWT`; its `KG`	4,065 t; 7.81 m	4,660 t; 7.78 m	4,446 t; 7.87 m
`D − T`; basic freeboard	4.09 m; 1.87 m	4.65 m; 2.14 m	4.37 m; 1.87 m
Ratios and `Fn` at `T`	pass	fail: `B/T` 3.30, `T/D` 0.59, `Fn` 0.228	fail: `B/T` 3.53, `T/D` 0.61
`GM` without ballast, departure / arrival	−0.24 / −0.39 m	0.58 / 0.46 m	2.04 / 1.96 m
Ballast for `GM` = 0.15 m, departure / arrival	740 / 1,020 t	none	none

	0.45, 7 × 4 × 14	0.50, 7 × 4 × 16	0.50, 8 × 4 × 14
Target 0.06 `B`, departure and arrival	1.32 m: not reached	1.32 m: not reached	1.47 m: reached
Roll period, departure / arrival	44.7 / 45.2 s	23.4 / 26.4 s	14.2 / 14.6 s
Homogeneous capacity, TEU of 14 t	528	564	596
The five checks	**all pass**	ratios fail	ratios fail

The acceptable design. Only the first candidate passes all five checks: `s_H` = 0.45, 7 rows × 4 tiers × 14 bays, 128.4 × 22.05 × 11.33 m, floating at 7.23 m with `C_B` = 0.615. For this brief it is the whole set of acceptable designs.

It is stable only with ballast: 740 t in the double bottom on departure and 1,020 t on arrival bring `GM` to 0.15 m. With all the ballast its double bottom can take, `GM` reaches only about 0.3 m, far below the target of 0.06 `B` = 1.32 m (7.5 Criteria). With `GM` at 0.15 m its roll period is long, about 45 s.

The other two need no ballast on departure or arrival, but they float too shallow for their proportions. With 8,893 t of deadweight, the wider hull draws 6.95 m and the longer one 6.68 m, so `B/T` is too large and `T/D` too small; the longer one is also too slow for its length. They are not stretched (6.6 Close the balance): the method rejects them.

Against the real ship. The design needs 6,148 kW by eq. (21), against the 6,000 kW installed on the reference ship. It draws 7.23 m against 7.9 m, for a hull 15.5 m longer and 2 m wider carrying the same deadweight.

Stability lets it carry 528 TEU of 14 t; the reference ship, 2 m narrower, is published with 410. For the design, stability, not space, limits the heavy cargo.

What remains before the design is accepted. Power it with Holtrop and Mennen and close the balance again (6.1 Power). Check the freeboard with the ICLL corrections, and replace the assumed range, crew and deckhouse with the owner's.

What the example shows.

- The sequence is mechanical once the inputs are fixed: split, targets, rows and tiers, bays, ratio check.
- Statistics from 8 ft 6 in ships underestimate the depth of a high-cube ship; the ratio check catches it.
- `k_U` drives the length: calibrate it, and report the sensitivity with your candidates.
- The deck count of eq. (12) is uniform along the ship; the sight line is not.
- In Steps 5 to 8 the weights fix the draught, and stability, not space, limits the TEU of 14 t.

III.2 Validation · the estimates against two built ships

The capacity estimates of Step 2 · Lay out the containers and Step 4 · Capacity on deck and visibility and the regressions of Step 1 · Split the TEU and set targets were run on the real grids of two built ships: the reference ship of I.5 and KMTC Seoul, a 2,540 TEU ship. They show which estimate to trust.

Reference ship (6 rows × 4 tiers in the holds, 11 hold bays; 8 rows × 4 tiers on deck, 13 deck bays; `C_B` 0.68 assumed):

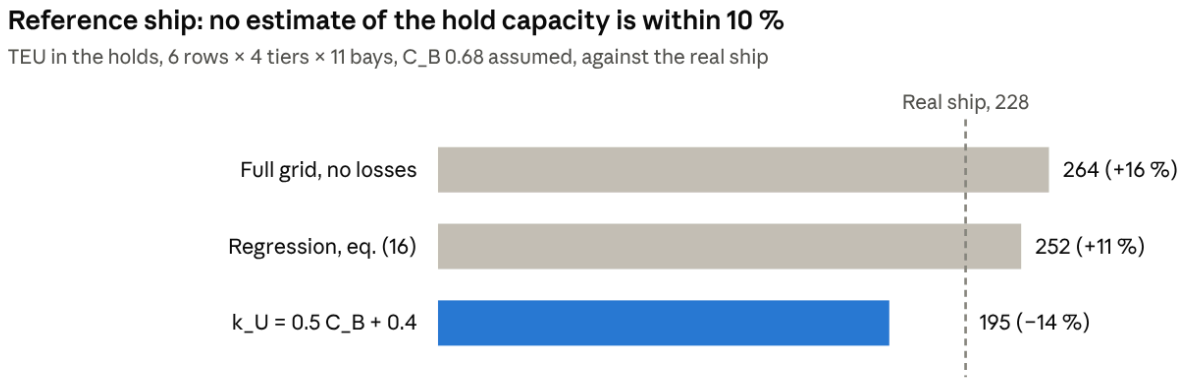


Figure III.3 · Reference ship: estimates of the TEU in the holds against the real ship; C_B 0.68 assumed

Estimate	Holds (TEU)	Deck (TEU)
Full grid, 11 · 6 · 4	264 (+16 %)	–
`k_U` formula, eq. (8), 264 × 0.74	195 (−14 %)	–
Regression, eq. (16), `p` = 24, `n_B` = 11	252 (+11 %)	–
Eq. (12), (11 + 2) · (6 + 2) · 4	–	416 (+13 %)
Real ship	228	368

The regressions of Step 1 · Split the TEU and set targets, entered with the ship's own `N_H` = 228, give `L_PP` 116.4 m, `B` 20.1 m, `D` 9.8 m and `T` 7.5 m, against 112.9, 20.0, 11.3 and 7.9 m. Length, beam and draught are within 6 %; depth is 13 % low, because the ship carries high cubes and the regressions come from ships with 8 ft 6 in containers.

KMTC Seoul (940 TEU in the holds, 1,600 on deck, 2,540 in total; `L_PP` 182 m, `B` 32.5 m, `w_DH` 2.15 m; Significant Ships of 2020): the regressions give `L_PP` 186.7 m, `B` 31.0 m and `D` 16.6 m (ship 16.8 m). From the target beam, eq. (7) gives ⌊(31.0 − 4.3) / 2.54⌋ = ⌊10.49⌋ = 10 rows; the ship has 11. Here the grid with one more row of Step 2 · Lay out the containers, 2.5 Bracket the targets: four grids per share is the real ship: rounding down loses half a row, which is exactly why the four grids are tried. With 11 rows, eq. (10) gives `B` = 32.2 m, 0.3 m less than the ship.

How to use the estimates.

- The regressions give `L`, `B` and `T` within a few percent; correct `D` for high cubes.
- The `k_U` formula is a first guess, about 14 % low on the reference ship. Calibrate it on a reference ship of your own size.
- Eq. (16) and the full grid both overestimate; use them to bound the answer, not to give it.
- Deck capacity is the least certain number: check it against the sight line and stability, not against a regression alone.
- Once the hull lines are drawn, the containers can be counted directly in the shape of the hull, which replaces all of these estimates.

Appendices

Appendix A · Nomenclature

Every symbol of the manual, with its unit, the step where it first appears, and the name used by Alvariño et al. (1997) where they have one, so you can find it in the source. Subscript 0 marks a target from statistics (I.4).

Capacity and the container grid

Symbol	Meaning	Unit	Alvariño et al.	First used
`N_TEU`	total capacity	TEU	NTEU	I.1, Step 1 · Split the TEU and set targets
`N_H`, `N_D`	TEU in the holds, on deck	TEU	NCHO, NCSC	I.1, Step 1 · Split the TEU and set targets
`s_H`	share of the TEU in the holds, `N_H / N_TEU`	–	–	I.1, Step 1 · Split the TEU and set targets
`n_B`, `n_R`, `n_T`	20 ft bays, rows and tiers in the holds	–	NCL, NCB, NCD	I.1, Step 2 · Lay out the containers
`n_HO`	holds, `n_B / 2`	–	NHOL	I.1, Step 3 · Main dimensions from the container grid
`n_l`	extra deck bays, for example over the engine room (0–2)	–	nl	Step 4 · Capacity on deck and visibility
`n_TD`	tiers on deck (at most 5)	–	nd	I.1, Step 4 · Capacity on deck and visibility
`k_U`	hold utilization: share of the full grid the ship really carries	–	CAPR	I.3, Step 2 · Lay out the containers
`p`	`n_R · n_T`, containers in the midship section of the hold, eq. (16)	–	–	Step 4 · Capacity on deck and visibility

Containers and allowances

Symbol	Meaning	Unit	Alvariño et al.	First used
`l_c`, `b_c`, `h_c`	container length (20 ft), width, height	m	LCON, BCON, DCON	I.1
`p_T`	tier pitch, `h_c + 0.013`	m	–	I.2, Step 7 · Stability
`l_hold`	length of one hold, eq. (6)	m	–	Step 2 · Lay out the containers
`g_LB`, `g_LF`	guide clearance lengthwise: at a bulkhead, free-standing	m	GLM, GLA	I.2
`g_BC`, `g_BF`	guide clearance across: at the coaming, free-standing	m	GBM, GBA	I.2
`w_DH`	double-hull width, each side	m	BDC	I.2
`h_DB`	double-bottom height	m	DDF	I.2
`h_CO`	hatch-coaming height	m	HES (DBR)	I.2

Symbol	Meaning	Unit	Alvariño et al.	First used
`c_HC`	margin under the hatch cover	m	HMAR	I.2
`c_DK`	deck camber	m	BRU	I.2
`t_cover`	hatch-cover thickness	m	–	I.2, Step 4 · Capacity on deck and visibility
`t_BH`	length taken by a transverse bulkhead	m	LMRO	Step 3 · Main dimensions from the container grid

Main dimensions and lengths

Symbol	Meaning	Unit	Alvariño et al.	First used
`L_PP`, `B`, `D`, `T`	length between perpendiculars, beam, depth, draught	m	LPP, B, D, T	I.2
`L_0`, `B_0`, `D_0`, `T_0`	targets from statistics; `T_0` is the freeboard draught, not a limit	m	–	I.4, Step 1 · Split the TEU and set targets
`L_OA`	length overall	m	–	I.2, Step 4 · Capacity on deck and visibility
`L_WL`	waterline length	m	–	Step 7 · Stability
`L_AP`, `L_ER`, `L_FP`	aft peak, engine room, fore peak	m	LAP, LER, LFP	I.2, Step 3 · Main dimensions from the container grid
`L_C`	cargo zone	m	LC	Step 3 · Main dimensions from the container grid

Visibility

Symbol	Meaning	Unit	Alvariño et al.	First used
`x`, `z`	position forward of the aft perpendicular; height above the base line	m	–	Step 4 · Capacity on deck and visibility
`x_eye`, `z_eye`	position and height of the eye on the bridge	m	–	Step 4 · Capacity on deck and visibility
`z_BR`	height of the navigation bridge above base, eq. (15)	m	–	Step 4 · Capacity on deck and visibility
`x_bow`, `x_fwd`, `x_sea`	forward end of the ship; forward end of a bay; where the sight line meets the sea	m	–	Step 4 · Capacity on deck and visibility
`z_base`	top of the hatch cover, where the first deck tier stands	m	–	Step 4 · Capacity on deck and visibility
`z(x)`	height of the sight line at `x`	m	–	Step 4 · Capacity on deck and visibility

Weights, form and power

Symbol	Meaning	Unit	Alvariño et al.	First used
`Δ`, `DWT`, `LWT`	displacement, deadweight, lightship	t	DISW, WPM, WR	I.4, Step 5 · Displacement, C_B and draught
`∇`	displaced volume	m^3	–	Step 7 · Stability
`C_B`, `C_M`, `C_P`, `C_WP`	block, midship, prismatic, waterplane coefficients	–	CB, CM, CP, CWP	Step 5 · Displacement, C_B and draught
`C_B'`	block coefficient corrected to 0.8 `D`, eq. (26)	–	–	Step 6 · Weights and power
`V`	service speed	kn	–	Step 5 · Displacement, C_B and draught
`Fn`	Froude number, `V / √(g L_PP)` with `V` in m/s	–	FN	Step 5 · Displacement, C_B and draught
`g`	acceleration of gravity, 9.81	m/s^2	–	Step 5 · Displacement, C_B and draught
`P`	installed power of the main engine, eqs. (21), (22) and (28)	kW	–	Step 6 · Weights and power
`P`	maximum continuous rating, eq. (17) only	BHP	–	Step 4 · Capacity on deck and visibility
`SFOC`	specific fuel oil consumption	g/kWh	–	Step 6 · Weights and power
`W_S`	steel weight	t	–	Step 6 · Weights and power
`E`	steel numeral, eq. (27)	m^2	–	Step 6 · Weights and power
`K`	steel weight coefficient, 0.036 for container ships	–	–	Step 6 · Weights and power
`l_1`, `h_1`; `l_2`, `h_2`	length and height of each superstructure; of each deckhouse	m	–	Step 6 · Weights and power

Weight groups

Symbol	Meaning	Unit	Alvariño et al.	First used
`C_N`	Admiralty coefficient, eq. (21)	$t^{2/3}$ kn^3/kW	–	Step 6 · Weights and power
`W_F`	fuel for the round voyage, eq. (22)	t	–	Step 6 · Weights and power
`R`	range of the round voyage, eq. (22); in eq. (29), the lightship margin	nm; t	–	Step 6 · Weights and power
`a_aux`	fuel of the auxiliary engines, as a share of the main engine's, eq. (22)	–	–	Step 6 · Weights and power
`C`	fuel reserve factor, eq. (22); in eq. (35), the roll-period factor	–	–	Step 6 · Weights and power
`W'_ST`, `C'_ST`	hull steel by Schneekluth, eq. (23); its coefficient	t; t/m^3	–	Step 6 · Weights and power

Symbol	Meaning	Unit	Alvariño et al.	First used
`∇_U`	volume under the deck plus the hatch coamings, eq. (23)	m^3	–	Step 6 · Weights and power
`C_BD`, `C_1`	block coefficient at the depth `D`; flare coefficient, eq. (23)	–	–	Step 6 · Weights and power
`KG'_ST`	height of the hull steel above base, eq. (24)	m	–	Step 6 · Weights and power
`D_S`	depth corrected for the deckhouse volume	m	–	6.6 Close the balance
`W_DH`, `C_DH`	deckhouse weight; volumetric coefficient of each layer, eq. (25)	t; kp/m^3	–	Step 6 · Weights and power
`l_DH`, `b_DH`, `h`	length and breadth of the deckhouse; height of one layer, eq. (25)	m	–	Step 6 · Weights and power
`k_1`, `k_2`, `k_3`	Müller–Köster corrections for layer height, internal walls and ship length, eq. (25)	–	–	Step 6 · Weights and power
`W_M`, `n`	machinery weight; engine speed, eq. (28)	t; rpm	–	Step 6 · Weights and power
`W_OT`, `K_OT`	outfit weight; outfit coefficient, eq. (29)	t; t/m^2	–	Step 6 · Weights and power

Stability

Symbol	Meaning	Unit	Alvariño et al.	First used
`z_hold,i`, `z_deck,i`	height of the centre of gravity of tier `i`, in the hold and on deck	m	–	Step 7 · Stability
`m_j`, `z_j`	mass and height of item `j`	t, m	–	Step 7 · Stability
`KG`, `KB`	height of the centre of gravity, of the centre of buoyancy, above base	m	–	Step 7 · Stability
`BM`, `GM`	metacentric radius, metacentric height	m	–	Step 7 · Stability
`I_T`	transverse moment of inertia of the waterplane	m^4	–	Step 7 · Stability
`δGM_FS`	free-surface correction	m	–	Step 7 · Stability
`T_φ`	natural roll period	s	–	Step 7 · Stability
`C`	roll-period factor of the IS Code, eq. (35)	–	–	Step 7 · Stability
`m_D`, `z̄_D`	mass and mean height of one more deck tier, eq. (36)	t, m	–	Step 7 · Stability

Free surface

Symbol	Meaning	Unit	Alvariño et al.	First used
`ρ_F`	density of the fuel in the slack tank, eq. (34)	t/m³	–	Step 7 · Stability
`l_t`, `b_t`	length and breadth of the slack tank, eq. (34)	m	–	Step 7 · Stability

Appendix B · Notes on the sources

Where this manual departs from its sources, it does so for the reasons below. Use the manual's version in the project.

Topic	Source says	Use in the project
Container height	The formulas of §3.8.6 assume 8 ft 6 in containers; the app. 2.9 regressions come from 8 ft 6 in ships; Papanikolaou Table 1.9 assumes 8 ft	high cubes, 2.896 m, unless the brief says otherwise; expect a deeper ship than the regressions give
Depth	Eq. 2.9.3 leaves out the camber and the 13 mm gaps; Eq. 3.8.31 includes both	Eq. 3.8.31, as eq. (11) (Step 3 · Main dimensions from the container grid)
Block coefficient	Papanikolaou Table 1.8 takes `C_B` from `L` and `Fn` (§2.10)	Katsoulis with `L`, `B`, `T` and `V`, solved with `Δ` (Step 5 · Displacement, C_B and draught)
Beam from rows	Papanikolaou Eq. 2.50, `B ≈ 3 n_R + 2.2` m	Eq. 2.9.2, as eq. (10) (Step 3 · Main dimensions from the container grid); Eq. 2.50 gives 35.2 m for 11 rows where KMTC Seoul has 32.5 m
New Panamax beam	49 m in older texts	51.25 m, the lock limit since 2018
Visibility distance	the program of §3.8.6.2 uses 2 `L_PP` from the forward perpendicular	SOLAS V/22: 2 `L_OA` or 500 m ahead of the bow, whichever is less (Step 4 · Capacity on deck and visibility)
Deck count	Eq. 2.9.5 puts the same number of tiers on every bay	apply the sight line bay by bay and subtract the tiers it cuts (Step 4 · Capacity on deck and visibility)
Hold regressions	Eqs. 3.8.28 and 3.8.29 meet at `L_PP` = 185 m	expect a jump of about 24 % across 185 m; use them only as a second opinion
Deck regression, Eq. 3.8.37	fitted on ships with the engine room aft	it can go negative for small ships; use eq. (12) and the sight line instead
`k_U = 0.5 C_B + 0.4`	for use without a better statistic	it reaches 0.9 at `C_B` = 1 and gives 0.74 where the reference ship has 0.86; calibrate it, and show the sensitivity (I.3, III.1)
Eye height	Eq. 3.8.36 gives the height of the bridge deck only	the eye 1.8 m above the bridge deck, an assumption of this manual (Step 4 · Capacity on deck and visibility)
Ship data cards	the data card of Cecilie Maersk gives 8 holds	text and drawing show 10 holds
Guide weights	a formula in the footnote of Table 2.9.2	not used: Schneekluth's steel formula already includes the cell guides (6.3 Steel)

Appendix C · Alvariño's boxes and the sections of this manual

Alvariño et al. number each calculation of their flow charts as a box, from 1 to 99. This manual numbers the same calculations by its own steps and sections (Figure II.1). Use this table only to find a calculation in their book.

Alvariño's box	What is calculated	Section of this manual
1	split the TEU in the holds into bays, rows and tiers	1.1 Split the TEU, 1.2 Targets from the TEU in the holds, 2.3 Rows and tiers to 2.5 Bracket the targets: four grids per share
2	length `L_PP`	3.1 Length
3	beam `B`	3.2 Beam
4	depth `D`	3.3 Depth
5	displacement	5.1 First displacement
6, 7	block coefficient and draught	5.2 Solve C_B and T
8, 9	`C_M` and `C_P`	5.3 The other coefficients
10, 11	`C_WP` and the centre of buoyancy	5.3 The other coefficients, from the course
12	freeboard check	3.4 Check the ratios and 5.4 Close the iteration with step 2
13	ratio check	3.4 Check the ratios and 5.4 Close the iteration with step 2
14	power	6.1 Power
15 to 21	deadweight items	6.2 Deadweight items
22	steel	6.3 Steel
23	machinery	6.4 Machinery
24	outfit	6.5 Outfit
25, 26	lightship and weight balance	6.6 Close the balance
27	capacity check	2.4 Bays and 4.1 Containers on deck
28 to 33	heights of the weights, `KG`	7.1 Height of the cargo and 7.2 KG of the ship
34 to 36	`KB`, `BM`, `GM`	7.3 KB, BM and GM
37	`GM` check	7.5 Criteria
38, 39	roll period and its check	7.6 Roll period
50 to 99	technical evaluation	after this manual

Appendix D · Writing the code

Your group writes its own program. This appendix says what the program must do, lists the functions it needs step by step, and gives the code of the harder steps in full. The other functions are each one formula of Part II, named in the card of its step; write them yourselves. Python 3 with no extra packages is enough.

What the program must do.

1. Hold the inputs of Step 0 in one block at the top: the brief (`N_TEU`, `V`, `DWT`), the container height, and the allowances and lengths of your reference ship.

2. Run Steps 1 to 4 for every share `s_H` from 0.30 to 0.50 in steps of 0.05, with the four grids of each (as rounded, one more tier, one more row, both): 20 candidates.
3. Keep the candidates that pass the ratio check of 3.4 and carry their deck TEU in at most five tiers.
4. Close the weight balance of each survivor by draught (Steps 5 and 6), load it in the conditions of 7.4 and find the ballast it needs (Step 7).
5. Apply the five checks of 8.1 and report the candidates that pass all five.
6. Print, for each candidate, the numbers the tables of III.1 show, so that a reader can follow every step.

Check it on the worked example first. With the inputs of III.1 your program must give the numbers of III.1: the sweep of Figure III.1, the three candidates and the accepted design. Only then run it on your own brief.

The functions, step by step. The names are those of the step cards; the equation numbers are those of this manual.

Step	Functions	What they do	Equations
1	`split`, `targets`	split `N_TEU` into `N_H` and `N_D`; first `L_0`, `B_0`, `D_0`, `T_0`	(1) to (3)
1.3	`length_from_trend`, `share_matching_trend`	cross-check with Fig. A.47	none
2	`rows`, `tiers`, `hold_utilization`, `bays`	rows and tiers rounded down; `k_U`; the smallest even number of bays	(7), (8)
3	`length`, `beam`, `depth`, `ratio_limits`, `check_ratios`	main dimensions from the grid; the ratio check	(9) to (11)
4	`deck_teu`, `deck_tiers_needed`, `bridge_height`, `tiers_visible`	deck TEU; tiers under the sight line	(12) to (15)
5	`cb_katsoulis`, `solve_cb_t`, `midship_coefficient`, `froude_number`	`C_B` and `T` together; `C_M`	(19), (20)
6	`admiralty_power`, `consumables`, `hull_volume`, `steel_schneekluth`, `steel_kg`, `deckhouse_weight`, `steel_weight`, `machinery_weight`, `lightship`, `close_weights`, `close_design`	power, deadweight items, lightship and the weight balance	(21) to (30)
7	`tier_heights`, `centre_of_gravity`, `load_cargo`, `stability`, `ballast_capacity`, `ballast_needed`, `homogeneous_capacity`	loading conditions, `GM`, roll period and ballast	(31) to (36)
8	`basic_freeboard`, `step8`	the five checks	none new
Sweep	`size_grid`	Steps 2 to 4 for one grid, one line of results	none

Code of the harder steps. Most first programs go wrong in four places: a geometry with two heights (the sight line), an iteration that must converge (`C_B` and `T`), a balance in which every weight depends on the draught, and a loading condition built from many weights. The code of those four is below. It uses the input names of Step 0 (`V`, `DWT`, `N_TEU`, `w_DH`, ...) and the constants of Steps 6 and 7 as global names, and calls functions of the table above that you write.

Step 4 · the tiers a deck bay can carry under the sight line, eqs. (13) and (14).

```
def tiers_visible(x_fwd, x_eye, z_eye, x_bow, T, distance, z_base, h_c):
    """Eqs. (13) and (14): tiers a deck bay can carry under the sight line.

    The sight line goes from the eye (x_eye, z_eye) to the sea surface at
    'distance' ahead of the bow (SOLAS V/22: 2 L_OA or 500 m, whichever is less).
    x is measured forward from the aft perpendicular, z up from the base line.
    x_fwd  : forward end of the bay, m
    T      : draught, m
    z_base : top of the hatch cover, where the first deck tier stands, m
    """
    x_sea = x_bow + distance
    z_line = T + (z_eye - T) * (x_sea - x_fwd) / (x_sea - x_eye)   # height of the line at x_fwd
    n = math.floor((z_line - z_base) / (h_c + gap))
    if n < 0:
        n = 0
    return n
```

Step 5 · `C_B` and the draught together, eq. (19).

```
def solve_cb_t(Delta, L_PP, B, V):
    """Eq. (19): find C_B and T together for a displacement Delta (t).

    Start from a draught, compute C_B with Katsoulis, then the draught that
    gives Delta = 1.030 C_B L_PP B T. Repeat until T changes by less than 1 cm.
    Returns C_B and T.
    """
    T = B / 2.9                                  # starting draught, any value works
    for i in range(50):
        C_B = cb_katsoulis(L_PP, B, T, V)
        T_new = Delta / (1.030 * C_B * L_PP * B)
        if abs(T_new - T) < 0.01:
            return C_B, T_new
        T = T_new
    print("The C_B and T loop did not converge")
    return None, None
```

Step 6 · the weight balance of eq. (30), closed by draught.

```
def close_design(s_H, n_R, n_T, show):
    """Steps 5 and 6 for one candidate: the weight balance of eq. (30), closed by draught.

    As in Lecture 3.1: lightship + deadweight against the hull. Here C_B follows the
    draught (Katsoulis, eq. 19) and the lightship follows T, C_B and the power, so the
    balance is repeated until the draught changes by less than 1 cm. Then k_U is
    recomputed with the new C_B (step 5.4); if the number of bays changes, it starts
    again. With show=True it prints each pass.
    """
    N_H, N_D = split(N_TEU, s_H)
    C_B = 0.68                                                  # starting value (step 2)
    for outer in range(10):
        n_B = bays(N_H, n_R, n_T, hold_utilization(C_B))
        L = length(n_B, L_AP_ER, L_FP)
        B = beam(n_R, w_DH, g_BC, g_BF)
        D = depth(n_T, h_DB, h_CO, c_HC, c_DK, h_c)
        n_HO = n_B // 2
        l_hold = 2 * 0.115 + 2 * l_c + 0.180                    # eq. (6), g_LB = 0.115, g_LF =
0.180
        T = B / 2.9                                             # starting draught
        for i in range(50):
            hull = 1.030 * C_B * L * B * T                      # what the hull floats now
            P = admiralty_power(hull, V)                        # power for that displacement
            ls = lightship(L, B, D, T, C_B, P, n_HO, l_hold)    # lightship from its parts
            shortfall = ls["LWT"] + DWT - hull                  # > 0: the hull is too small
            Delta, C_B, T_new = close_weights(DWT, ls["LWT"], L, B, V)   # eq. (30): new C_B and T
            if show:
                print(f"    pass {i + 1}: T {T:.2f} m, LWT + DWT {ls['LWT'] + DWT:.0f} t,"
```

```
                      f" hull {hull:.0f} t, shortfall {shortfall:+.0f} t (reserve {ls['R']:.0f} t)"
                      f" -> T {T_new:.2f} m, C_B {C_B:.3f}")
            if abs(T_new - T) < 0.01:
                T = T_new
                break
            T = T_new
        if bays(N_H, n_R, n_T, hold_utilization(C_B)) == n_B:
            break
    Delta = 1.030 * C_B * L * B * T
    return {"s_H": s_H, "N_H": N_H, "N_D": N_D, "n_R": n_R, "n_T": n_T, "n_B": n_B,
            "L": L, "B": B, "D": D, "T": T, "C_B": C_B, "Delta": Delta,
            "P": admiralty_power(Delta, V), "ls": ls}
```

Step 7 · one loading condition, and the least ballast that gives `GM` ≥ 0.15 m.

```
def stability(c, condition, ballast, n_teu):
    """Step 7 for one candidate c (from close_design): one loading condition,
    n_teu containers and a mass of water ballast in the double bottom, t.

    condition: "departure", "arrival" or "homogeneous" (section 7.4).
    Returns T, KG, KB, BM, the free-surface correction, GM and the roll period.
    """
    L, B, D, T, C_B = c["L"], c["B"], c["D"], c["T"], c["C_B"]
    cons = consumables(c["P"], V)
    tanks = cons["fuel"] + cons["lub"] + cons["water"]
    people = cons["food"] + cons["crew"]
    if condition == "arrival":
        tanks = 0.10 * tanks
        people = 0.10 * cons["food"] + cons["crew"]
    mass = m_homog if condition == "homogeneous" else m_design
    n_TD = deck_tiers_needed(c["N_D"], c["n_B"], n_l, c["n_R"])
    per_deck_tier = (c["n_B"] + n_l) * (c["n_R"] + 2)
    cargo_m, cargo_z = load_cargo(n_teu, mass, c["N_H"], c["n_T"], n_TD, per_deck_tier, D)
    W_DH, z_DH, h_layer = deckhouse_weight(L, B, D)
    masses = [c["ls"]["LWT"], tanks, people, ballast] + cargo_m
    heights = [c["ls"]["KG_L"], z_tanks, z_DH, z_tanks] + cargo_z
    KG, Delta_c = centre_of_gravity(masses, heights)
    # hydrostatics at the draught of this condition (wall-sided, as Demo 5)
    C_WP = (1 + 2 * C_B) / 3                                  # eq. (34), Papanikolaou Eq. 2.83
    T_c = T + (Delta_c - c["Delta"]) / (1.030 * C_WP * L * B)
    C_Bc = Delta_c / (1.030 * L * B * T_c)
    KB = T_c * (5 / 6 - C_Bc / (3 * C_WP))                    # eq. (33)
    C_IT = C_WP * (0.1272 + 0.8724 * C_WP)                    # eq. (34), Papanikolaou Eq. 2.95
    BM = C_IT * L * B**3 / 12 / (C_Bc * L * B * T_c)          # eq. (33)
    l_tank = 2 * 0.115 + 2 * l_c + 0.180                      # one hold long
    b_tank = (B - 2 * w_DH) / 2                               # half the inner bottom
    dGM_FS = rho_fuel * l_tank * b_tank**3 / 12 / Delta_c      # eq. (34), one slack tank (ASSUMED)
    GM = KB + BM - KG - dGM_FS                                # eq. (33)
    C = 0.373 + 0.023 * B / T_c - 0.043 * L / 100             # eq. (35), L_WL taken as L_PP
    T_roll = 2 * C * B / math.sqrt(GM) if GM > 0 else None
    return {"T": T_c, "Delta": Delta_c, "n_teu": n_teu, "ballast": ballast, "KG": KG,
            "KB": KB, "BM": BM, "dGM_FS": dGM_FS, "GM": GM, "T_roll": T_roll}


def ballast_needed(c, condition, n_teu):
    """Section 7.4: the least double-bottom ballast (in steps of 10 t) that gives
    GM >= 0.15 m. Returns the stability result, or None if the double bottom
    cannot give that GM."""
    ballast = 0.0
    while ballast <= ballast_capacity(c, condition, n_teu):
        st = stability(c, condition, ballast, n_teu)
        if st["GM"] >= GM_min:
            return st
        ballast = ballast + 10
    return None
```

Appendix E · References

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